Master Geomechanics, Civil Engineering and Risks

Numerical study of the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a stochastic poro-mechanical approach

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Finally, I would like to thank my family and friends for their constant encouragement and support throughout my studies. Their confidence in me has been a source of motivation during both the rewarding and challenging moments of this journey.

Declaration on the Use of AI-Assisted Technologies

In preparing this dissertation, the author used generative AI tools (Gemini and Claude) for limited technical assistance only: support with coding and debugging (CAST3M scripts and PYTHON post-processing), language editing and structure suggestions, and help in quickly organising and synthesising data during the research process. The scientific content, the analyses, the interpretation of the results and the conclusions presented in this work were produced by the author.

No AI tool was used to generate scientific conclusions, to fabricate or alter data, or to replace the author's own critical analysis. All AI-assisted material was reviewed, verified and corrected where necessary, and integrated under the author's responsibility. The author retains full responsibility for the content of this dissertation.

Abstract

Concrete spalling, the sudden explosive detachment of concrete surface when concrete is exposed to rapidly rising temperature, is a major threat to the integrity of structures subjected to fire or to severe accident conditions, such as nuclear containment buildings. It results from coupled thermo-hydro-mechanical (THM) process of the material, which is still incompletely understood and remains difficult to predict.

The work studies the THM behaviors of concrete under severe accident conditions using a stochastic poro-mechanical approach. A coupled THM model is implemented in the finite element code CAST3M. In which the thermo-hydric subsystem is solved monolithically, and the mechanical problem (linear elasticity with a MAZARS damage law) is coupled in a staggered manner.

Unlike the traditional approach of TH coupling in [5], this work adopts the new thermo-hydro-chemical (THC) approach of [13]. The transport and storage properties are redefined using the *hydration degree*, this ensures that the initial hydro-chemical state of the material is integrated into the model.

The model is then calibrated and validated against the experiment conducted by Kalifa [10], which served as a reference benchmark, with the aim of reproducing the measured temperature and gas pressure evolutions within concrete specimens. The temperature field is reproduced accurately, whereas the gas pressure cannot be matched with the same confidence. This raises the question of whether the gauges truly recorded the pore gas pressure.

Finally, the outcomes of the calibrated model contribute as the input for a stochastic assessment of the model, carried out with OPENTURNS. Therefore, it provides insights of the correlation and sensitivity analyses that identify the dominant parameters; a reliability analysis quantifies the probability of failure and the risk of spalling; and a random-field description of material heterogeneity is shown to produce more critical predictions than the homogeneous assumption.

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Chapter 1

Introduction

1.1 Context and problem statement

Concrete structures exposed to extreme heat undergo a complex interpaly of thermal, hyric, and mechanical phenomena that con lead to progressive degradation, and in most severe cases, to *spalling*, the sudden, explosive detachment of concrete layers from the exposed surface.

This phenomenon has been a critical concern for civil infrastructure such as nuclear containment buildings and tunnels, where the consequences of spalling as the loss of cover and cross-section can damage the load-bearing capacity and functions of the structure.

Recently, its catastrophic consequences are also relevent in aerospace engineering. In 2023, a SpaceX rocket test has failed due to the destructive power of concrete spalling [12]. As the rocket ignited on its concrete launch pad, the immense heat generated triggered explosive spalling. The debris seen flying around the launch site consisted of sharp concrete flakes. These flakes shot upwards, severely damaged the test field and the rocket itself, ultimately causing the rocket test launch to fail.

(a) The crater in the launchpad after SpaceX's Starship took off.

(b) Debris litters the Starship launchpad, with damaged fuel tanks visible in the background.

Figure 1.1: Catastrophic consequences of explosive concrete spalling during the SpaceX Starship test flight.

1.1.1 Concrete behaviour under high temperature

When concrete is heated rapidly, several coupled physical processes occur simultaneously

- **Thermal:** Steep temperature gradients develop throughout the concrete's cross-section, causing differential thermal expansion and generating compressive stresses at the heated surface.
- **Hydric:** Both free and chemically bound water evaporate and move through the pore network. In low-permeability concretes such as high-performance concrete (HPC), vapor cannot escape fast enough, resulting in pore pressure build-up behind a quasi-saturated *moisture clog*.
- **Mechanical:** The combined effects of thermal stresses and pore pressure build-up on the solid skeleton can exceed the tensile strength, which is degraded by temperature. This initiate cracks parallel to the heated surface and potentially lead to explosive spalling.

These three fields are not independent: they interact through a network of coupling mechanisms that any predictive model must reproduce. Despite extensive experimental work, the mechanisms governing spalling are still incompletely understood and no single, universally accepted theory has emerged [5].

1.1.2 The initial hydro-chemical state of concrete

The response of concrete to fire does not depend only on the heat: it also depends strongly on the state of the material *before* heating. The calcium–silicate–hydrates (C–S–H) formed during cement hydration control the strength, the porosity and the amount of chemically bound water that can later be released by dehydration.

The degree to which hydration has progressed, together with the residual moisture content inherited from curing and drying, therefore conditions the transport and storage properties that govern pore pressure build-up. Accounting for this initial hydro-chemical state is the key idea behind recent thermo-hygro-chemical (THC) models [13], and it motivates the formulation adopted in this work.

1.1.3 Limitations of existing models

Existing THM models present several limitations that this work seeks to address:

- The initial hydro-chemical state is often treated implicitly, through empirical functions of temperature only, rather than through an explicit hydration-degree description;
- The uncertainty in the material parameters is rarely assessed: properties are typically taken as homogeneous, deterministic values, so the influence of their variability on the predicted response is left unquantified.

1.2 Objectives and outline of the report

Main objectives of this internship are structured in the remainder of the report as follow:

- Chapter 2 presents the coupled THM model of concrete at high temperature, from the governing equations and coupling mechanisms to the monolithic thermo-hydric matrix system and the staggered mechanical step, and details the hydration-degree reformulation following [13] and extending the framework of [5].
- Chapter 3 describes the reference experiment of [10] and its numerical implementation in Cast3M (geometry, mesh, material data, thermal loading and curing/drying conditions).
- Chapter 4 reports the calibration of the temperature and gas-pressure responses, validates the model against the experimental and numerical references, and discusses the limits of the gas-pressure calibration.
- Chapter 5 performs a stochastic assessment of the calibrated model with OpenTURNS: propagating the uncertainty in the material parameters through random sampling to rank the influential parameters (Sobol sensitivity analysis); quantify

the probability of failure and the risk of spalling through reliability analysis and with a complementary look at the effect of spatial heterogeneity.

- Chapter 6 summarizes the main findings and outlines perspectives for future work.

1.3 Research enviroment and Computational tools

This work was carried out at the *Laboratoire Sols, Solides, Structures – Risques* (3SR), Université Grenoble Alpes, which provided the scientific environment and the expertise in multiphysics modelling and risk analysis.

The coupled THM model is implemented in Cast3M, an open-source finite-element code developed at CEA. Compared with closed multiphysics platforms, COMSOL Multiphysics, Cast3M offers greater transparency and control for implemnting tightly coupled THM while COMSOL offers a more user-friendly GUI.

The uncertainty quantification relies on OpenTURNS, an open-source library based on Python used to generate the random samples, propagate them through the model and compute the Sobol indices and reliability.

Finally, Python (in Visual Studio Code) is also used to automate the calibration workflow and post-process the results, and LaTeX to typeset this report.

Chapter 2

Theoretical framework

[REVIEW] The old content of this chapter is mostly moved to Appendix A. This chapter is a draft version and its outline is as follow.

2.1 Thermal and hydric phenomena in heated concrete

When one face of a concrete member is exposed to a rising temperature, a sequence of strongly interacting thermal and hydric processes develops through the thickness. They are summarised here; the full constitutive relations are derived in Appendix **??**.

Thermal phenomena. Heat is transmitted into the material essentially by *conduction*, governed by Fourier's law, while *convection* and *radiation* act on the heated boundary and are accounted for through the boundary conditions. Energy transport is not purely conductive: the bulk flow of liquid water and gas advects enthalpy, and the phase changes (evaporation, dehydration) consume energy and locally slow the temperature rise. The resulting energy balance and the temperature dependence of the heat capacity ρC_p and conductivity λ are detailed in Appendix **??**.

Hydric phenomena. Concrete is treated as a multiphase porous medium made of a solid skeleton (s), liquid water (l), water vapour (v) and dry air (a). On heating, free water *evaporates* and chemically bound water is released from the calcium–silicate–hydrates (C–S–H) through *dehydration*. The generated vapour is transported by pressure-driven *Darcy flow* and by *Fickian diffusion* within the gas mixture. Part of the vapour migrates inward, reaches cooler regions and *recondenses*, building a quasi-saturated layer — the *moisture clog* — that blocks further gas transport. Behind this clog the gas pressure p_g rises sharply (Section A.5.2). The governing transport and mass-balance relations are given in Appendix **??**.

These two families of phenomena are the physical origin of the two spalling mechanisms described next.

2.2 Spalling mechanisms

Spalling is the violent or non-violent detachment of layers or pieces of concrete from the surface of a heated structural element. It can cause premature structural failure through reinforcement exposure and cross-section reduction, as illustrated by several catastrophic tunnel fires (Channel Tunnel 1996, Mont-Blanc 1999, St. Gotthard 2001) [**thmframework**]. Despite extensive experimental work, no single theory has reached consensus; the two most widely accepted mechanisms, and their combined action, are presented below.

2.2.1 Thermal-gradient mechanism

A steep temperature gradient through the thickness produces differential thermal dilation:

1. **Compressive stresses in the heated cover**: the surface layers tend to expand but are restrained by the cooler interior, generating compressive stresses parallel to the heated face.
2. **Tensile stresses in the interior**: the restraint of the expanding surface induces tension in the deeper layers.
3. **Fracture and energy release**: when the tensile stress exceeds the tensile strength f_t, cracks form parallel to the surface; the sudden release of stored elastic energy can drive the explosive ejection of fragments.

High-performance concrete (HPC) is particularly prone to this mechanism because its higher brittleness stores more elastic energy before failure, leading to a more violent release.

2.2.2 Pore-pressure mechanism (moisture clog)

The sequence of events is:

1. **Dry zone**: near the heated face, free water evaporates and bound water is released by dehydration, forming a dry zone.

2. **Vapour migration**: part of the vapour escapes towards the heated face, part is driven inward.

3. **Condensation / moisture clog**: the inward vapour condenses where $T < T_{sat}$, forming a quasi-saturated, nearly impermeable barrier.

4. **Pressure build-up**: behind the clog, p_g rises sharply as evaporation feeds a region from which vapour cannot escape; thermal dilation of liquid water in saturated pores adds to the build-up.

5. **Tensile failure**: when p_g exceeds the tensile strength, cracks form parallel to the surface, potentially leading to explosive spalling.

The governing parameters are the intrinsic permeability K_0 (low permeability hinders vapour evacuation) and the initial moisture content (more water to evaporate), both promoting higher pore pressures.

2.2.3 Combined mechanism

In practice, explosive spalling generally results from the *combined action* of pore pressure and thermal stress. The spalling condition can be written schematically as

$$\sigma_{thermal}(T) + \sigma_{pore}(p_g, p_c, S_l) \geq f_t(T), \tag{2.1}$$

where $\sigma_{thermal}$ is the thermal stress from differential dilation, σ_{pore} the stress from pore-pressure loading (via the effective stress, Section 2.5), and $f_t(T)$ the temperature-degraded tensile strength. Equation (2.1) makes clear why a complete coupled model is needed: neither a purely thermal nor a purely hydric analysis can capture the full picture.

2.3 Thermo-hydric (TH) coupled model

The thermal and hydric fields are described by three primary variables,

$$\mathcal{U}_{TH} = \{\, p_g,\ p_c,\ T \,\}, \tag{2.2}$$

the gas pressure, the capillary pressure and the temperature. Each is associated with a conservation equation. After substitution of Darcy's law, Fick's law and the constitutive relations (Appendix **??**), the final coupled forms read:

Dry-air conservation.

$$\begin{aligned}
&\phi(1-S_l)\left(\frac{\partial\rho_a}{\partial T}\frac{\partial T}{\partial t}+\frac{\partial\rho_a}{\partial p_c}\frac{\partial p_c}{\partial t}+\frac{\partial\rho_a}{\partial p_g}\frac{\partial p_g}{\partial t}\right)+(1-S_l)\rho_a\frac{\partial\phi}{\partial T}\frac{\partial T}{\partial t}\\
&-\phi\rho_a\left(\frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t}+\frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t}\right)-\nabla\cdot\left(\frac{K\rho_a k_{rg}}{\mu_g}\nabla p_g\right)-\nabla\cdot\left(D\rho_g\frac{M_vM_a}{M_g^2}\nabla\frac{p_a}{p_g}\right)=0.
\end{aligned}\tag{2.3}$$

Water-species conservation.

$$\begin{aligned}
&\big(S_l\rho_l-(1-S_l)\rho_v\big)\frac{\partial\phi}{\partial T}\frac{\partial T}{\partial t}+S_l\phi\frac{\partial\rho_l}{\partial T}\frac{\partial T}{\partial t}+\phi(\rho_l-\rho_v)\left(\frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t}+\frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t}\right)\\
&+(1-S_l)\phi\left(\frac{\partial\rho_v}{\partial T}\frac{\partial T}{\partial t}+\frac{\partial\rho_v}{\partial p_c}\frac{\partial p_c}{\partial t}+\frac{\partial\rho_v}{\partial p_g}\frac{\partial p_g}{\partial t}\right)-\nabla\cdot\left(\frac{K\rho_l k_{rl}}{\mu_l}(\nabla p_g-\nabla p_c)\right)\\
&-\nabla\cdot\left(\frac{K\rho_v k_{rg}}{\mu_g}\nabla p_g\right)-\nabla\cdot\left(D\rho_g\frac{M_vM_a}{M_g^2}\nabla\frac{p_v}{p_g}\right)=-\dot m_{dehyd}.
\end{aligned}\tag{2.4}$$

Energy conservation.

$$\rho C_p\frac{\partial T}{\partial t}-K\left(C_l\frac{\rho_l k_{rl}}{\mu_l}(\nabla p_g-\nabla p_c)+C_g\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g\right)\cdot\nabla T-\nabla\cdot(\lambda\cdot\nabla T)=-H_{vap}\dot m_{vap}+H_{dehyd}\dot m_{dehyd}.\tag{2.5}$$

2.3.1 TH coupling mechanisms

Equations (2.3)–(2.5) are *two-way coupled*:

- **Temperature drives the hydric field.** It enters through *mass source terms* — evaporation $\dot m_{vap}$ and dehydration $\dot m_{dehyd}$ are thermally activated — and through *temperature-dependent transport coefficients*: permeability $K(T)$, porosity $\phi(T)$, viscosities $\mu(T)$, saturation vapour pressure $p_{vs}(T)$ and diffusivity $D(T)$.
- **The hydric field drives the temperature.** The bulk flow of water and gas *advects* enthalpy (the advection term in Eq. (2.5)), while phase changes act as heat sinks ($-H_{vap}\dot m_{vap}$, $+H_{dehyd}\dot m_{dehyd}$), and the moisture content modifies ρC_p and λ.

The momentum balance and the resulting mechanical response are introduced in Section 2.5.

2.4 Improved thermo-hydro-chemical (THC) model

2.4.1 Motivation

In the TH model above, dehydration enters only as a prescribed source term $\dot{m}_{dehyd}(T)$ and the porosity follows a simple linear law $\phi = \phi_0 + A_\phi(T - T_0)$ (Eq. **??** in Appendix **??**). This treatment ignores three facts that strongly influence spalling: (i) the *chemical state* of the cement paste before the accident depends on its hydration/curing history; (ii) the release of chemically bound water from the C–S–H is the chemical process that controls porosity, permeability and strength at high temperature; and (iii) chemical damage is essentially *irreversible*. The improved model adopted here, following Sciumè, Moreira and Dal Pont [13], resolves the chemical field explicitly and couples it to the thermal and hydric fields.

2.5 Coupling to the mechanical field

The mechanical response is driven by the converged TH/THC fields through a *one-way* (staggered) coupling: at each time step the fields (T, p_g, p_c) enter the mechanical problem as prescribed loading, and the resulting damage $\mathcal{D}$ feeds back weakly to the transport properties via Eq. (**??**) at the next step. The TH/THC fields act through two routes:

- **Free strains.** The total strain is decomposed as $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_\sigma + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{sh}$, where $\boldsymbol{\varepsilon}_{th}$ is the thermal dilation (driven by T) and $\boldsymbol{\varepsilon}_{sh}$ the capillary shrinkage (driven by p_c). Creep is neglected, so $\boldsymbol{\varepsilon}_\sigma = \boldsymbol{\varepsilon}_{el}$.
- **Pore-pressure loading.** Through Bishop's effective stress $\boldsymbol{\sigma}' = \boldsymbol{\sigma}^{Total} + \alpha\, p_s\, \mathbf{I}$, the average pore pressure $p_s = p_g - S_l p_c$ enters the load vector as an external loading; its build-up shifts $\boldsymbol{\sigma}'$ towards tension and promotes cracking when $f_t(T)$ is exceeded.

Together with the spalling criterion (2.1), this closes the physical loop between the transport and mechanical fields. The constitutive model (linear elasticity coupled with Mazars continuum damage [**mazars1986**]), its fracture-energy regularisation [**hillerborg1976**], and the spatial/temporal discretisation of the coupled problem are reported in full in Appendix **??**; only the elements needed to interpret the results of the following chapters are retained here.

Chapter 3

Experimental Reference and Numerical Model

[REVIEW] This chapter is a draft version and its outline is as follow.

3.1 The Kalifa experiment

[REVIEW] This section is currently a draft version

Purpose. The campaign of Kalifa et al. [10] quantifies the *thermo-hydral* process responsible for spalling. When concrete is heated on one face, free and chemically bound water are released as vapour; part migrates inward, recondenses in cooler regions and forms a quasi-saturated *moisture clog* that blocks gas evacuation and triggers a sharp pore-pressure build-up behind the drying front. The test measures *simultaneously and continuously* the temperature, the gas (pore) pressure at several depths, and the global mass loss, providing a quantitative dataset for validating coupled codes.

Set-up and instrumentation. A prismatic specimen of $30 \times 30 \times 12$ cm^3 is heated on one 30×30 cm^2 face by a radiant heater (up to 5 kW, 800 °C) placed 3 cm above the surface. The lateral faces are insulated with porous ceramic blocks, so heat and mass transfer are essentially one-dimensional. The specimen rests on a balance recording its mass. Five gauges measure pressure and temperature at 10 mm, 20 mm, 30 mm, 40 mm and 50 mm from the heated face, and a sixth tube measures the surface temperature at 2 mm depth.

Concrete mixes. Two calcareous-aggregate mixes were tested: an ordinary concrete (OC, M30, $w/c = 0.5$) and a high-performance concrete (HPC, M100, $\approx 90\,\mathrm{MPa}$, $w/c = 0.34$). Specimens were sealed for about six months, giving a homogeneous moisture state. The low w/c and high compactness of the HPC strongly reduce its intrinsic permeability, the main parameter controlling the pressure amplitude.

Heating scenario and measured data. The heater was set to 450 °C, 600 °C and 800 °C as a step signal. The 600 °C test (duplicated for repeatability) is retained as the reference, since it produces a well-developed pressurisation/relief cycle while preserving the 1-D condition. The measured response shows a temperature rise with a short plateau near 200 °C, a two-stage gas-pressure history (pressurisation then relief as the front passes), and a continuous mass loss. In HPC the pore pressure exceeds the saturating vapour pressure $p_{vs}(T)$, confirming the contribution of enclosed air and of the thermal dilation of liquid water. These three quantities are the validation targets.

3.2 Numerical model

[REVIEW] This section is currently a draft version

Geometry and mesh.

Material parameters.

Thermal input and boundary conditions.

Curing, drying and initial conditions. A key feature of the THC framework [13] is that the hygro-chemical state before the accident is obtained from a preliminary *curing* simulation rather than prescribed arbitrarily. Starting from

$$T_0 = 20\,^\circ\mathrm{C}, \quad p_{g,0} = 101\,325\,\mathrm{Pa}, \quad \mathrm{RH}_0 = 0.5, \quad \Gamma_0 = 0, \tag{3.1}$$

hydration develops over the curing duration, establishing self-consistent profiles of hydration degree Γ, saturation S_l and relative humidity. The converged curing fields are then used as initial conditions for the heating (drying) stage, ensuring that the initial water content and hydration state — which strongly control the pressure amplitude — are physically consistent with the specimen history.

Noted that $\mathrm{RH}_0 = 0.5$ in curing duration reflect a inrealistic intial conditions, but prevent the code from crashing. An attempt to calibrate p_g results by reverting this value to a high as $\mathrm{RH}_0 = 0.98$ is done but the outcome show that this value is not useful in the calibration stage Section 4.3.

Solution strategy and numerical robustness. At high temperature the THC system is *strongly nonlinear*: the liquid–vapour phase change and the sorption law vary abruptly around 100 °C; dehydration $F(T)$ and the Arrhenius hydration term add stiff, temperature-activated source terms; and the permeability and porosity vary by orders of magnitude across the drying front. The Newton iterations may therefore *diverge*, particularly when the moisture clog and the steep front form. Three measures are investigated to restore convergence:

- **Increasing the time-integration parameter θ.** Moving the θ-scheme toward a fully implicit integration ($\theta \to 1$), combined with under-relaxation of the Newton updates, damps the oscillations of the nonlinear coefficients and stabilises the iteration.
- **Reducing the time step.** Adaptive time stepping that cuts Δt when the residual stalls resolves the fast transients (front passage, pressure peak) and enlarges the convergence radius of Newton's method.
- **Refining the mesh.** A finer discretisation near the heated face reduces the inter-element jumps in T, S_l and the transport coefficients, avoiding the spurious oscillations that trigger divergence.

3.3 Application of the THC model

3.3.1 Computational Tool: A Customized Cast3M Framework

The numerical simulations in this study were conducted using a highly customized version of the open-source finite element software Cast3M (base version 2024). Unlike standard mechanical or thermal problems, modeling the Thermo-Hydro-Mechanical (THM) behavior of High-Performance Concrete (HPC) at high temperatures requires handling highly non-linear, multiphysics phenomena. To address these challenges, the standard Cast3M environment was enriched with advanced pre-release procedures and monolithic solvers developed by Sciumè and Dal Pont [13].

The necessity of utilizing this custom-compiled version—executed directly from the source directory rather than the standard binary distribution—stems from several critical computational limitations of the vanilla software:

- **Complex Boundary Conditions:** In standard Cast3M, imposing convective mass flux (such as water vapor escaping the heated concrete surface) alongside Dirichlet or Neumann thermal boundary conditions is not natively straightforward. Core resolution procedures, such as `TRANSNON`, had to be deeply modified to accurately simulate the mass exchange between the concrete surface and the ambient environment during the drying and accident phases [5].

- **Monolithic TH Solver Implementation:** To overcome the severe convergence issues and high computational costs associated with traditional staggered (partitioned) approaches, this custom version employs a fully coupled monolithic strategy for the Thermo-Hydric (TH) equations. This allows the primary variables (temperature T, gas pressure P_g, and capillary pressure P_c) to be solved simultaneously at each node within the same time loop.

- **Integration of the THC Approach:** The custom build inherently integrates specific macro-procedures (e.g., `@BET_THM`, `@POWERS`, and `@SATURA`) required to evaluate the state variables. Specifically, it allows the intrinsic permeability and porosity to evolve dynamically as a function of the *effective hydration/dehydration degree* rather than purely empirical temperature-dependent functions.

Chapter 4

Model Calibration and Validation

4.1 Model calibration

The accuracy of TH/THC model relies on a large number of material parameters – intrinsic permeability, thermal conductivity and heat capacity, sorption properties, surface exchange coefficients and a complex set of constitutive laws describing their evolution with temperature and dehydration.

Many critical properties at high-temperature such as the evolution of intrinsic permeability, sorption isotherms are extremely difficult or even impossible to measure directly through experimental techniques and the values often contain uncertainties due to the intrusive nature of the sensors. Furthermore these are not measured directly in the Kalifa campaign.

A pre-determined parameter set therefore cannot be expected to reproduce the experiment. Consequently, a calibration procedure for the numerical models is strictly required to identify, within physically admissible ranges, the parameter set that makes the model reproduce the measured temperature and gas-pressure histories. Only once this reference case is calibrated and validated can the model be trusted for uncertainty studies in Chapter 5.

4.1.1 Calibration strategy

At the first approximation, the thermal problem is weakly coupled to the hydric one: the temperatur field drives evaporation, dehydration and the gas-pressure build-up, while the effect of the hydric state on the temperature field is relatively small within the experimental range. This hierachy promotes a two-stage calibration process:

1. **Stage 1 – Temperature.** The temperature evolution is calibrated independently

of the pressure field. In this stage, the *surface* (boundary) parameters should be calibrated first, then the *bulk* conduction parameters.

2. **Stage 2 – Gas pressure.** With the reliably calibrated temperature field as a fixed thermal loading, the hydric parameters gorvening the gas-pressure evolution are adjusted, from the most influential to the secondary ones.

The calibration targets are the temperature and gas-pressure histories measured at 10, 20, 30, 40 and 50 mm from the heated face, together with the near-surface temperature recorded at 2 mm (denoted the 0 mm measurement).

4.1.2 Automated parameter sweep

The sweeps are driven by a Python script that launches Cast3M repeatedly, one run per parameter combination, without manual intervention. For each stage the parameter values to be tried are listed explicitly, and `itertools.product` builds every possible combination of those lists; a single loop then feeds each combination to Cast3M in turn.

This automation relies on two supporting pieces of code.

- First, the Cast3M input code itself exports the results of each run to a `.csv` file – the temperature and gas-pressure histories at the measurement points.
- Second, a separate Python post-processing script reads these `.csv` files and plots the computed histories against the experimental measurements [10] and also against the results from Dorjan Dauti's thesis [5]. All combinations then can be visually compared, and the best-fit set is selected by the author based on this comparison.

For example, in the gas-pressure calibration stage, the two parameters swept are the intrinsic permeability K_0 (denoted `KINT`) and the permeability–hydration coefficient A_Γ (denoted `AK`), with the trial values:

- $K_0 \in \{1{\times}10^{-19},\ 2{\times}10^{-19},\ 1{\times}10^{-20},\ 5{\times}10^{-20},\ 7.5{\times}10^{-20},\ 8.5{\times}10^{-20}\}\ \mathrm{m}^2$ (6 values);
- $A_\Gamma \in \{3,\ 4,\ 5,\ 6,\ 7,\ 8\}$ (6 values).

These two lists give $6 \times 6 = 36$ parameter pairs, i.e. 36 Cast3M runs. The same driver is reused by simply changing the value lists.

In every stage, a wider range of each parameter is first explored to bracket the admissible values; the search is then narrowed to the few representative values reported in the following sections, which are the ones shown in the figures.

[REVIEW] The following sections are currently under review and need to be rewritten

4.2 Temperature calibration

4.2.1 Surface temperature (00mm): boundary heat exchange coefficient

The first quantity calibrated is the temperature of the heated face. In the experiment, heat is supplied by a radiant heater on the top-hot face, so the surface energy balance is mostly influneced by radiation and convection. See Equation (A.2) and Equation (A.4):

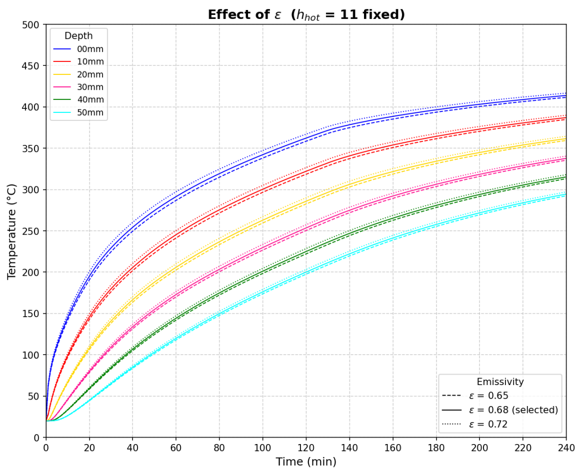
$$q_{00\,\mathrm{mm}} = h^{\mathrm{hot}}\,(T_\infty - T_s) + \varepsilon\,\sigma\left(T_{\mathrm{sur}}^4 - T_s^4\right), \tag{4.1}$$

where h^{hot} is the convective heat-transfer coefficient at the top-hot surface, ε the surface emissivity (radiative coefficient), σ the Stefan–Boltzmann constant, T_{sur} and T_∞ are the surrounding temperature provided by the radiant heat source and T_s the surface temperature.

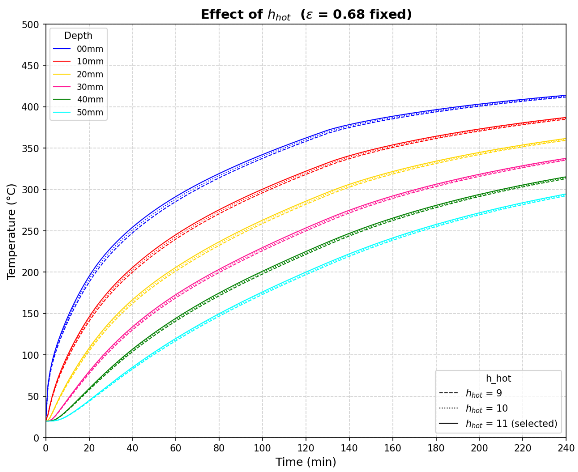
The two boundary coefficients h^{hot} and ε are adjusted so that the computed 0 mm history matches the measured near-surface (2 mm) response. At this point the conduction parameters are held at their reference values; only the energy input at the surface is tuned.

The surface response is calibrated over three values of ε (0.65, 0.68, 0.72) and three of h_c (9, 10, 11 $\mathrm{W\,m^{-2}K^{-1}}$).

Because the resulting curves differ only slightly, the full 3×3 grid is not informative; instead the effect of each parameter is shown one at a time, varying it while the other is held at its retained value (Fig. 4.1). The pair retained at this stage is $\varepsilon = 0.68$ and $h^{\mathrm{hot}} = 11\ \mathrm{W\,m^{-2}K^{-1}}$.



(a) Effect of ε ($h_c = 11$ fixed).



(b) Effect of h_c ($\varepsilon = 0.68$ fixed).

Figure 4.1: Surface (0 mm) temperature: sensitivity to the emissivity ε (a) and to the convective coefficient h_c (b), compared with the measured near-surface response.

Two distinct effects are concluded from Figure 4.1.

- The emissivity ε is the most sensitive parameter: because the radiative term acts in a strong non-linear way as T^4 in Equation (4.1), a small increase in ε raises the surface temperature substantially, and mostly in the high-temperature part of the curve, where radiation dominates the heat input.
- The convective coefficient h^{hot} acts more gently and mainly on the *early* part of the curve: increasing h^{hot} raises the initial heating rate (the linear $h^{\mathrm{hot}}(T_\infty - T_s)$ term is largest while T_s is still low), with only a slight additional rise in the tail.

In brief, ε sets the level of the high-temperature plateau while h^{hot} controls the initial ramp; tuning the two together reproduces the measured surface temperature evolution.

4.2.2 In-depth temperature: thermal conductivity and heat capacity

Once the surface temperature is reproduced, the temperature rise at depth (10–50 mm) is governed by heat conduction through the material, i.e. by the volumetric heat capacity ρC_p (Equation (A.7)) and the thermal conductivity λ (Equation (A.10) and Equation (A.11)), both given in Appendix A. Most of the quantities in these laws (ϕ, S_l, the phase densities ρ and the temperature T) are imposed by the state of the material. The two genuinely adjustable inputs are:

- the solid specific heat C_{ps} (denoted `CS` in the model), which enters the heat capacity through Eq. A.7. Although a temperature dependence is defined in Appendix A, in the present implementation it is kept *constant* – it is not updated with temperature;
- the reference dry conductivity λ_{d0} (denoted `KS` in the model), which scales the whole conductivity law.

A subtlety arises here: adjusting the surface condition in Section 4.2.1 changes the energy that actually enters the specimen, which in turn shifts the in-depth histories. Consequently λ_{d0} and C_{ps} cannot be calibrated in isolation – they must be *re-calibrated* jointly with the surface condition.

The calibration therefore proceeds as a short journey, starting from the parameter set carried over from the surface stage and adjusting one quantity at a time.

Starting point. The run is first launched with the reference set inherited from the surface calibration in Section 4.2.1: $\varepsilon = 0.68$, $h^{\mathrm{hot}} = 11\ \mathrm{W\,m^{-2}K^{-1}}$, $C_{ps} = 948\ \mathrm{J\,kg^{-1}K^{-1}}$

and $\lambda_{d0} = 1.67\ \mathrm{W\,m^{-1}K^{-1}}$. Compared with the experiment and with the earlier results of Dauti (Figure 4.2), the temperature is already well reproduced near the surface, but the deeper probes (30, 40 and 50 mm) are over-predicted: the in-depth temperatures are too high and still need to be calibrated.

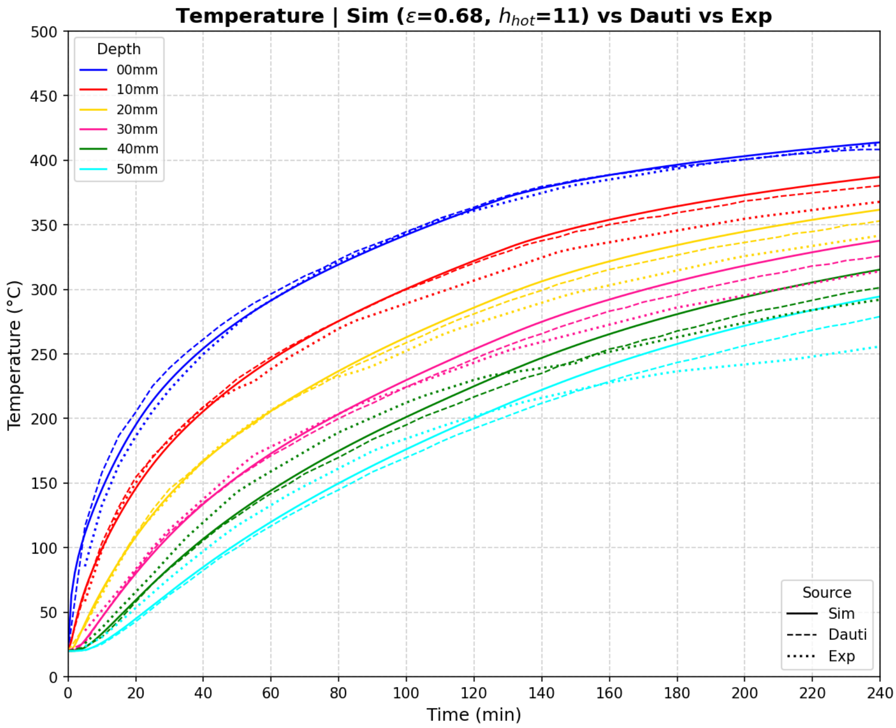


Figure 4.2: Temperature with the reference set ($\varepsilon = 0.68$, $h_c = 11$, $C_{ps} = 948$, $\lambda_{d0} = 1.67$), compared with the experiment and with Dauti's results.

Step 1 – reducing the conductivity λ_{d0} (`KS`). The conductivity acts mostly on the layers *closer to the heated face*. To bring down the over-predicted 10 and 20 mm curves, λ_{d0} is reduced (Figure 4.3). This indeed lowers those curves, but it has a side effect: by slowing the inward transfer, heat accumulates near the surface, so the surface temperature becomes too high in the tail of the curve and, for the lowest values, the computation eventually crashes.

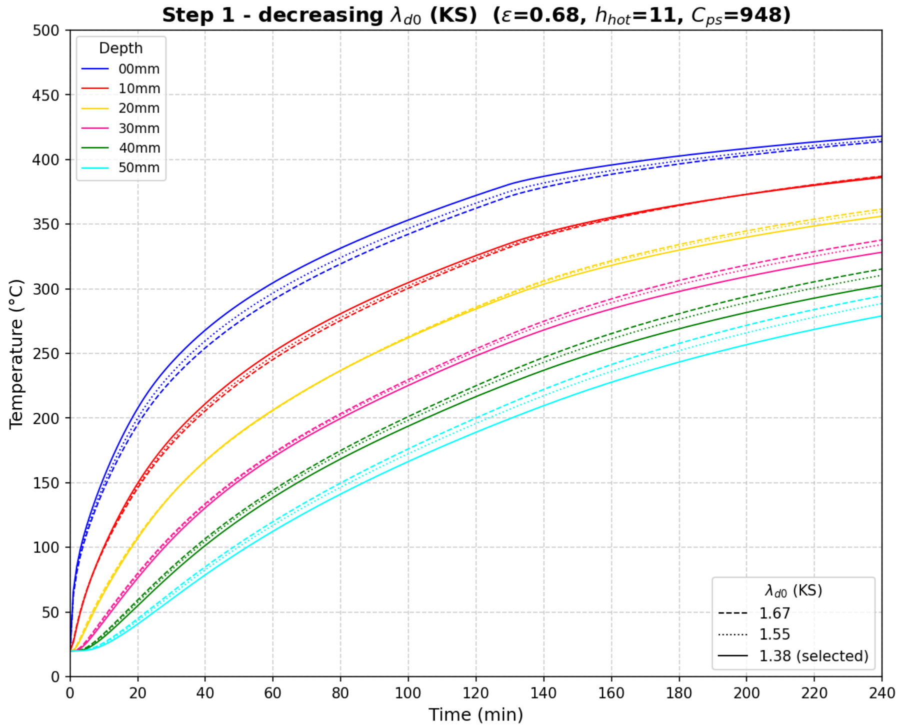


Figure 4.3: Effect of decreasing the conductivity λ_{d0} (KS)

Step 2 – re-calibrating the emissivity ε. The surface overheating introduced by the lower λ_{d0} is corrected by re-calibrating the emissivity. Reducing ε lowers the radiative input at the surface (Fig. 4.4) and restores the 0 mm response,stabilising the run.

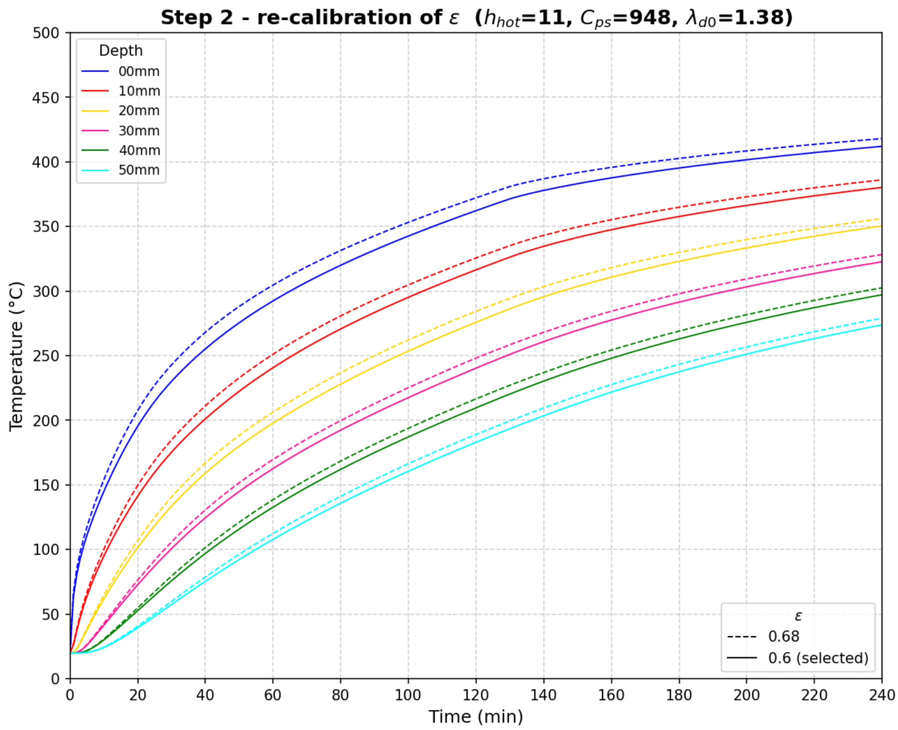


Figure 4.4: Re-calibration of the emissivity ε after the conductivity was reduced

Step 3 – increasing the heat capacity C_{ps} (CS). The heat capacity governs the thermal inertia and is felt mostly at the *deepest points*. To bring down the still over-predicted 30, 40 and 50 mm curves, C_{ps} is increased (Fig. 4.5): a larger C_{ps} stores more energy per unit temperature rise, so the same heat flux produces a smaller, more delayed temperature increase in the core, lowering the internal temperatures.

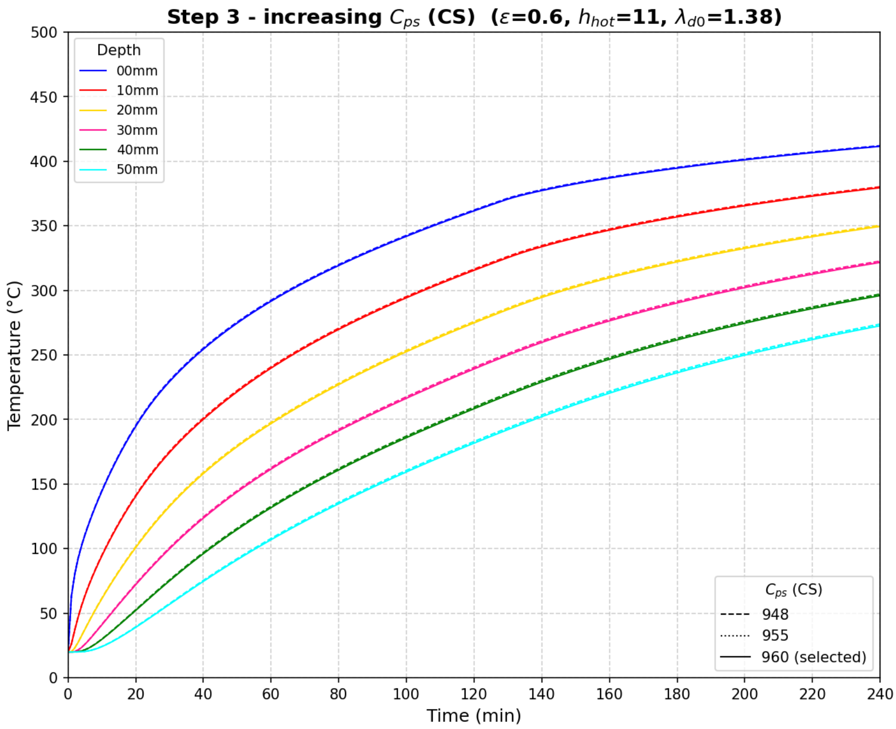


Figure 4.5: Effect of increasing the heat capacity C_{ps} (CS):

Balancing C_{ps} and λ_{d0}. Increasing C_{ps}, however, cannot be pushed independently of λ_{d0}: the two parameters are coupled and must be balanced against each other. If C_{ps} is raised while λ_{d0} is kept low, the material both stores a large amount of energy and conducts it inward too slowly, so the incoming heat cannot diffuse away from the surface. This was observed directly for the combination $\varepsilon = 0.68$, $h^{\text{hot}} = 11$, $C_{ps} = 960$, $\lambda_{d0} = 1.38$: the surface heat could not pass inward, the surface temperature rose abnormally, and the run crashed. The combination only becomes stable once the surface input is reduced – by lowering the emissivity to $\varepsilon = 0.6$ (Step 2) – so that a high C_{ps} and a low λ_{d0} can coexist. The calibration therefore consists of finding a balanced (λ_{d0}, C_{ps}) pair, together with the matching surface input, rather than tuning either conduction parameter on its own.

At the end of this journey the temperature is reproduced consistently at all depths (0–50 mm). The final comparison against the Dauti and experimental curves is shown in Figure 4.6, providing a reliable thermal loading for the gas-pressure calibration.

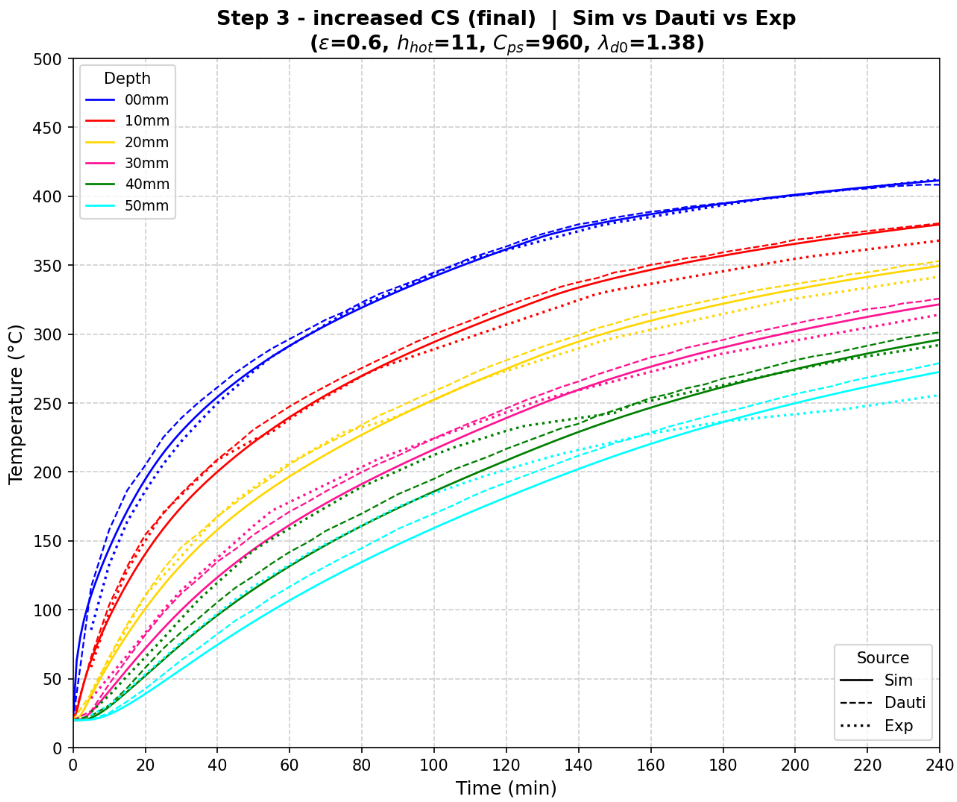


Figure 4.6: Temperature with the calibrated set ($\varepsilon = 0.6$, $h_c = 11$, $C_{ps} = 960$, $\lambda_{d0} = 1.38$): simulation (solid) vs. Dauti (dashed) vs. experiment (dotted) at 10–50 mm.

4.3 Gas-pressure calibration

With the temperature field fixed, the calibration of the gas-pressure evolution is addressed. The pressure build-up is governed primarily by how easily vapor can escape the heated region (permeability) and the amount of water available for vaporization.

4.3.1 Primary parameters: K_0 and A_Γ

The intrinsic permeability K_0 (denoted `KINT`) and the permeability–hydration coefficient A_Γ (denoted `AK`) are the two most influential parameters dictating the magnitude and acceleration of the gas transport. In the newly implemented THC model, the evolution of permeability is highly sensitive to the degradation of the effective hydration degree $\tilde{\Gamma}$??.

These two parameters must be strictly balanced. If the combination makes K too low, vapour cannot escape and the trapped vapor generates extremely severe pressure gradients; if it makes K too high, the vapor escapes too easily into the environment, resulting in flattened, underestimated pressure profiles. In both cases, the Newton-Raphson solver diverges (crashes), so K_0 and A_Γ have to be calibrated carefully together.

Starting point – $K_0 = 2 \times 10^{-19}$ m^2. The first sweep is performed at the initial reference permeability $K_0 = 2 \times 10^{-19}$ m^2, varying A_Γ over 3, 4 and 5 (with $A_\Gamma = 3$ as the initial reference); the result is shown in Figure 4.7. Increasing A_Γ raises the peaks, but this K_0 value is not physically representative of HPC, and the curves still do not fit the experimental pressures.

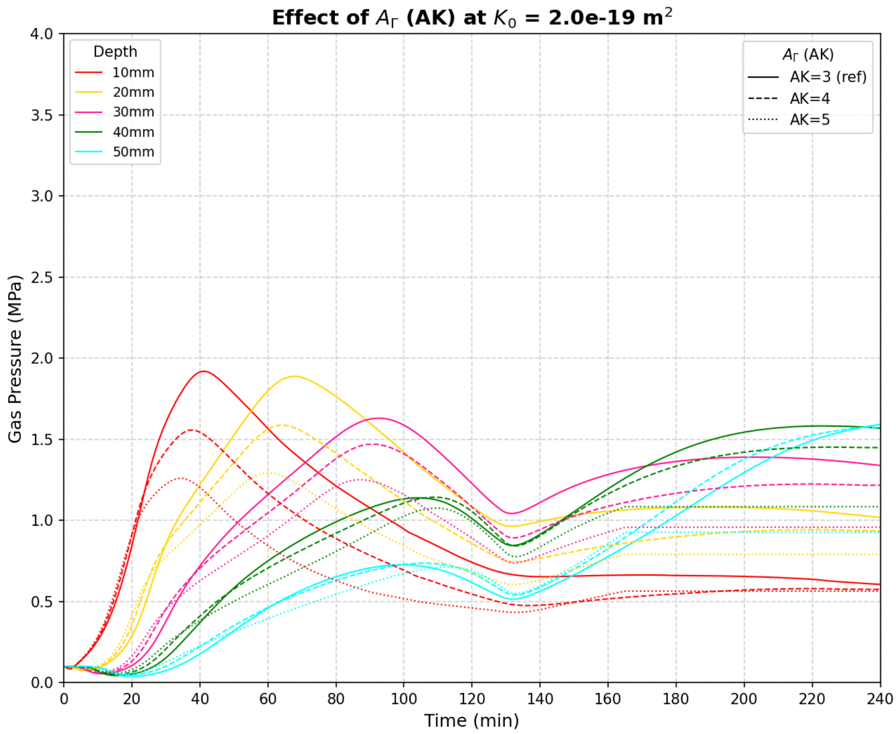


Figure 4.7: Effect of A_Γ ($A_K = 3, 4, 5$) at the initial reference permeability $K_0 = 2 \times 10^{-19}$ m^2

Reducing the permeability – $K_0 = 8.5 \times 10^{-20}$ m^2. The permeability is therefore reduced to $K_0 = 8.5 \times 10^{-20}$ m^2,which is more consistent with a dense HPC, and the A_Γ sweep is repeated Figure 4.8) to assess the effect of lowering K_0. Trapping the gas more effectively raises the peaks, and the pair $K_0 = 8.5 \times 10^{-20}$ m^2, $A_\Gamma = 4$ is retained as the best compromise.

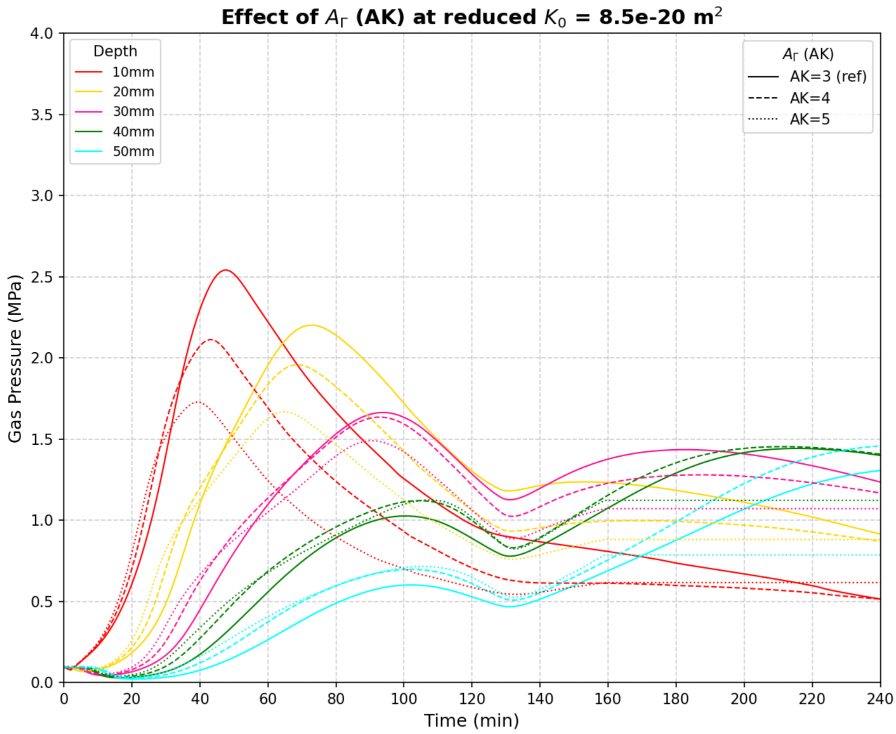


Figure 4.8: Effect of A_Γ ($A_K = 3, 4, 5$) at the reduced permeability $K_0 = 8.5 \times 10^{-20}$ m^2

Comparison with the references. The retained pair is compared against the Dauti and experimental curves in Figure 4.9. The agreement is reasonable in the outer layer of 10 mm, but a structural difficulty appears with depth, as discussed below.

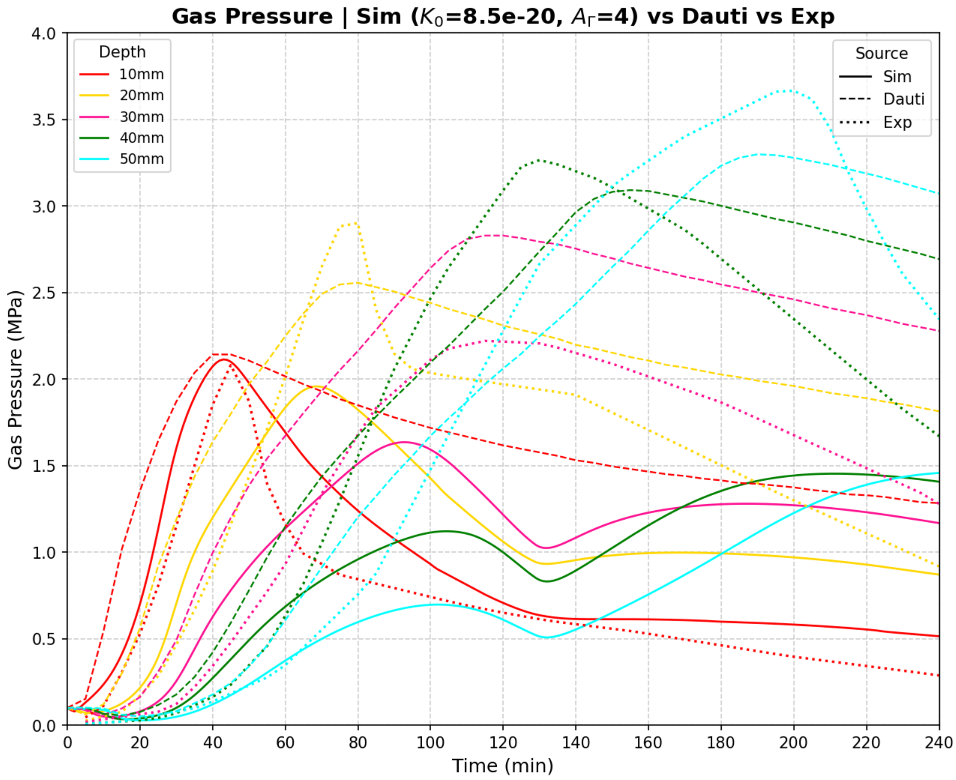


Figure 4.9: Gas pressure with the retained pair ($K_0 = 8.5 \times 10^{-20}$ m^2, $A_\Gamma = 4$): simulation (solid) vs. Dauti (dashed) vs. experiment (dotted)

The Main Issue: The sweep over (K_0, A_Γ) reveals the central difficulty of this stage: the gas pressure can be adjusted significantly only in the *outer layers* (10 and 20 mm). As one moves deeper into the specimen (30, 40, 50 mm), the simulated peak pressure *decreases*. This contradicts the experimental observations of Kalifa et al. [10], where deeper zones exhibited progressively higher peak pressures.

Physically, this discrepancy suggests that the high-gas-pressure region generated by the simulation is spatially smaller than it should be. The pressure build-up and the formation of the "moisture clog" remain concentrated too close to the exposed surface, failing to propagate massive condensation and pressurization into the deeper layers of the concrete core.

4.3.2 Secondary adjustments investigated

To address this depth-dependent pressure decay and force the moisture clog deeper into the material, several secondary parameters were investigated:

- **Initial Relative Humidity (RH_0):** While intuitively a higher initial saturation provides a larger water reservoir for vaporization, increasing RH_0 towards full saturation (e.g., > 0.99) induces severe numerical singularities in the capillary pressure gradients, systematically causing the code to crash.

- **Desorption Isotherm (a):** Adopting a stiffer desorption curve (higher a parameter, which is physically more appropriate for HPC to retain water at higher capillary pressures) successfully yielded a slight increase in the peak pressure at 20 mm, but failed to propagate this increment to the $30 - 50$ mm depths.

- **Aging / Curing time:** Modifying the hydration history significantly alters the initial state. Decreasing the aging time reduces the overall gas pressure and critically compromises the code's stability. On the other hand, increasing the aging time excessively subjects the HPC to severe self-desiccation, rendering the internal core unrealistically dry prior to heating.

- **Vapor mass-transfer coefficient (h_g):** In an attempt to artificially trap the gas, the convective exchange coefficient at the heated face was reduced to a minimum. While this approach successfully forced vapor inward and increased p_g in the deeper zones, it could only maintain numerical stability if compensated by an artificially large intrinsic permeability (K_0). However, imposing a large K_0 fundamentally contradicts the extremely dense, low-permeability nature of HPC containing silica fume.

The outcome of the last attempt is shown in Figure 4.10: with $h_g = 0.001$ and a large $K_0 = 2 \times 10^{-18}$ m^2, increasing A_Γ (1 to 4) does raise the deeper-zone pressures, but the overall pressure level is depressed because such an abnormally large K_0 lets the gas escape too easily.

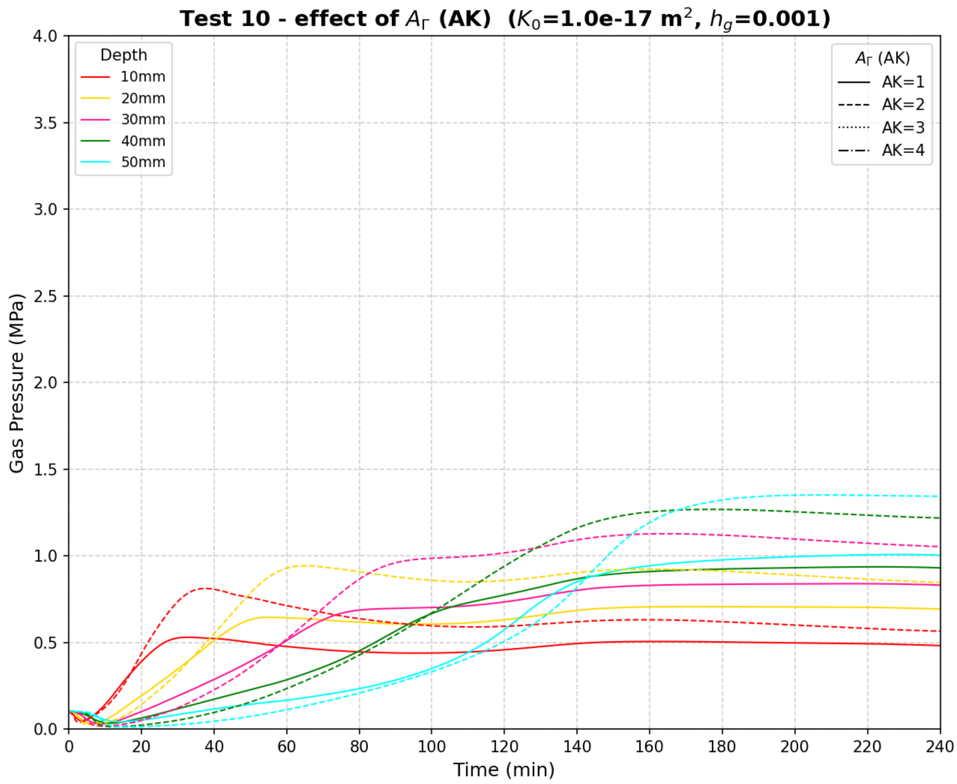


Figure 4.10: Gas-pressure histories at 10–50 mm for a minimal vapour transfer coefficient $h_g = 0.001$ and a large $K_0 = 2 \times 10^{-18}$ m^2, sweeping A_Γ from 1 to 4.

In summary, none of these secondary adjustments – alone or in combination with K_0 and A_Γ – could perfectly reproduce the deep pressure peaks without violating either the physical constraints of the material or the mathematical stability of the solver.

4.4 Validation and discussion

The reference model utilizes the calibrated parameter set optimized primarily for the thermal field and the near-surface gas pressure. Based on the calibration journey, the following critical discussions and validation outcomes are drawn.

Table 4.1: Final calibrated parameter set retained for the reference model.

Stage	Parameter	Retained value
Surface heat exchange	Convective coefficient h_c	11 $\mathrm{W\,m^{-2}K^{-1}}$
	Emissivity ε	0.6
Bulk conduction	Reference dry conductivity λ_{d0}	1.38 $\mathrm{W\,m^{-1}K^{-1}}$
	Solid specific heat C_{ps}	960 $\mathrm{J\,kg^{-1}K^{-1}}$
Gas pressure	Intrinsic permeability K_0	8.5×10^{-20} m^2
	Permeability–hydration coeff. A_Γ	4

4.4.1 Temperature

The temperature validation is the in-depth comparison already obtained at the end of the calibration (Figure 4.6): with the calibrated set of Table 4.1, the model reproduces the measured temperature histories at all depths within a few degrees, confirming that the thermal sub-problem and its boundary conditions are correctly captured. Importantly, the gas-pressure parameters K_0 and A_Γ calibrated in the last stage have a negligible effect on the temperature field, so the temperature agreement of Figure 4.6 remains valid for the final parameter set. The temperature field can therefore be used as a trustworthy loading for the hydric and mechanical sub-problems.

4.4.2 Gas pressure

The gas-pressure validation is the comparison obtained with the retained pair (Figure 4.9). The response **could not be validated**: even with the calibrated set of Table 4.1, and across the explored parameter space – K_0, A_Γ, RH_0, h_g, the desorption parameter a and the aging time – the model reproduces the pressure only in the outer layers (10 and 20 mm). In the deeper zones the simulated peak pressure keeps decreasing with depth, while the measurements show the opposite trend, so the high-pressure region remains too concentrated near the exposed surface.

Additionally, the conduction parameters C_{ps} and λ_{d0} retained for the temperature have a negligible effect on the gas pressure. With the fact K_0 and A_Γ ave a negligible effect on the temperature field, these justify the sequential calibration adopted in this chapter.

4.4.3 Dicussions

Two complementary explanations are proposed for this discrepancy.

- The first concerns the experiment itself. Measuring pore pressure requires inserting a gauge into the concrete, which inevitably creates a local discontinuity – a small cavity surrounded by an already porous and heterogeneous material. It is therefore not certain that these gauges record the true gas pressure of the intact porous network; part of what they measure may reflect the local conditions of the cavity and the surrounding damaged zone rather than the field the model computes. This raises a genuine question about what the deep-zone gauges are actually measuring, and hence about the reference against which the model is being judged.

- The second concerns the model. The formulation used here is the recent THC model, which is still being developed; its saturation, permeability and dehydration laws have not yet been fully tuned to the high-temperature behaviour of HPC.

> The fact that the deeper-zone pressures could be raised only through physically unreasonable parameter values (very high K_0, minimal h_g) suggests that the model is missing a mechanism that redistributes the pressure build-up further inward, and that further refinement of the THC formulation is required before a quantitative validation of the gas pressure can be expected.

Given the inherent uncertainties in both the non-linear material parameters and the ambiguous experimental gas-pressure data, relying solely on deterministic calibration is insufficient. This robustly justifies the necessity of shifting toward a probabilistic framework, utilizing uncertainty quantification to assess the spalling risk, which will be the core focus of Chapter 5.

Chapter 5

Stochastic assessment of the model

5.1 Sources of uncertainty and concrete heterogeneity

5.1.1 Motivation for probabilistic assessment

In civil engineering, a structure is considered safe when it remains fit for its intended use throughout its service life [2]. Deterministic models have traditionally been used to evaluate structural safety. However, the gap between a real physical system and a simplified numerical model, along with the unforeseeable nature of future loads and environmental conditions, inherently leads to uncertainty [1].

To manage these risks, modern engineering codes such as the Eurocodes adopt a probabilistic format [2]. This allows the safety of a structure to be explicitly quantified through its probability of failure P_f (Appendix B.5) and its reliability $R = 1 - P_f$. By introducing random variables into mechanical models and propagating them through structural analyses (Section 5.2), objective decisions can be made based on both the probability of an event and the severity of its consequences [1].

5.1.2 Classification of uncertainties

According to reliability theory, uncertainties can be classified into two categories [2, 1]:

- **Aleatoric uncertainties** arise from the natural randomness of the environment and the spatial or temporal variability of material properties. They cannot be reduced; they can only be quantified and accounted for in the design process [1].
- **Epistemic uncertainties** stem from incomplete knowledge or insufficient data. They include *model uncertainties* (discrepancy between mathematical models and

actual physical processes) and *statistical uncertainties* (limited or inaccurate data). Unlike aleatoric uncertainties, they can be reduced by improving models and gathering more data [1].

5.1.3 Concrete heterogeneity and material variability

A critical source of aleatoric uncertainty is the inherent heterogeneity of concrete [1, 6]. Its mechanical properties depend on the formation history, casting conditions, internal chemical reactions, and long-term degradation processes [6].

This variability manifests as *continuous* spatial fluctuations (e.g. Young's modulus, permeability, porosity) and *discrete* defects (e.g. micro-cracks, capillary networks) [1]. Consequently, material parameters must be modelled as random variables or random fields rather than deterministic values (Section 5.2).

5.2 Stochastic finite element framework

In structural engineering, material properties such as the Young's modulus, permeability, or thermal conductivity are never perfectly known. As noted by [6], even under strict quality control the compressive strength of high-performance concrete can range from 62 to 100 MPa. Two approaches exist to handle this uncertainty:

1. A *deterministic* analysis assigns a single fixed value to each parameter and produces a unique structural response; it cannot, by itself, quantify the likelihood that a design limit will be exceeded.

2. A *stochastic* analysis, on the other hand, treats uncertain parameters as random quantities, propagates this uncertainty through the model, and delivers statistical information—mean, variance, probability of failure—about the structural response.

Two levels of description are used in this work; their full definitions, moments and simulation methods are given in Appendix B.1 and summarised in Table B.1.

Random variable. Each uncertain property is described by a single distribution (PDF/CDF, mean, standard deviation, CoV; Appendix B.1.1) applied uniformly over the whole specimen in a given realisation. This is the level used for the sensitivity (Section 5.5) and reliability (Section 5.6) analyses; the uncertainty is propagated either by Latin Hypercube sampling (Section 5.4) or by Gaussian quadrature (Appendix B.3).

Random field. The property varies from point to point, subject to a spatial correlation structure characterised by a correlation length L_c (Appendix B.1.2). Realisations are generated by the Turning Bands Method and propagated by Monte Carlo (Appendix B.2); this level is used for the spatial-heterogeneity study of Section 5.7.

A random variable lets strong zones compensate weak ones, whereas a random field allows local weak zones to govern failure — the distinction examined in Section 5.7.

[REVIEW] The following sections are currently under review and need to be rewritten

5.3 Uncertain inputs and quantity of interest

5.3.1 Two separate analyses

The uncertainty is propagated through *two independent studies* rather than a single combined one:

Analysis A — intrinsic permeability K_0. A *direct* uncertainty on a transport property, identified in Chapter 4 as the dominant control on the gas-pressure magnitude.

Analysis B — mix composition. An *upstream* uncertainty on the mix-design proportions, introduced at its physical source and propagated through Powers' model (Chapter 2, Appendix A), which converts the proportions into porosity, hydration degree, sorption and the resulting transport properties.

The two are kept apart for two reasons. First, they are of different nature — a material-property scatter on one side, an upstream design imprecision on the other — and mixing them would blur the interpretation. Second, K_0 is known from Chapter 4 to dominate the pressure response; pooling it with the four mix proportions in a single variance decomposition would let its variance swamp the Sobol indices and *mask* the relative importance *among* the mix proportions. Separating the analyses lets each source be characterised on its own terms.

5.3.2 Choice of probability distributions

Three distribution families are used (Table 5.1), each chosen to reflect the *nature* of the corresponding uncertainty rather than by convenience; the formal definitions are given in Appendix B.1.1.

Lognormal — permeability K_0. The intrinsic permeability is strictly positive and varies by multiplicative factors: it is known only to within an order of magnitude and its scatter is naturally expressed as a ratio, not an additive offset. A lognormal law guarantees $K_0 > 0$ for every sample and reproduces this multiplicative, positively skewed variability; it is the standard model for the permeability of concrete [6, 15]. It is also consistent with the random-field generator of Section 5.7: the Cast3M `ALEA` operator produces a Gaussian field that is exponentiated into a lognormal one (Appendix B.2.3), so the random-variable and random-field descriptions of K_0 share the same marginal.

Normal — cement content c and aggregate content agg. The batched masses (in $\mathrm{kg\,m^{-3}}$) are controlled by the mixing plant around a target value. Their variability is the accumulation of many small, independent dosing and weighing errors, which by the central-limit theorem tends to a Gaussian centred on the nominal dosage. The scatter is small relative to the mean (CoV $\approx 4\%$ for c, $\approx 3\%$ for agg), so the symmetric Normal law describes the batch-to-batch dispersion faithfully and the probability of a physically meaningless (negative) sample is negligible.

Uniform — ratios w/c and s/c. For the water/cement and silica/cement ratios, what is known is not a measured central tendency but an *admissible design window*: an HPC of this type is specified to lie within $w/c \in [0.30, 0.38]$ and $s/c \in [0.08, 0.12]$. When only the bounds are known and there is no basis to prefer any value inside them, the maximum-entropy (least-informative) choice is the uniform distribution [1]. It models the *epistemic* imprecision of the as-specified mix rather than a measurement scatter, and it keeps every sample strictly within the physically admissible window, avoiding tails that would push Powers' model or the solver outside its valid range.

The four mix proportions are taken as mutually independent design choices; their physical coupling (a given w/c and s/c jointly set porosity and permeability) is produced *inside* the model by Powers' stoichiometry, not imposed on the inputs. The OpenTURNS declaration is given in Table 5.1.

Table 5.1: Random inputs of the two analyses. calibrated K_0 and the mix proportions

Input	Symbol	Distribution	Parameters
Analysis A — transport property			
Intrinsic permeability	K_0	Lognormal	$\mu = 8.5 \times 10^{-20}\ \mathrm{m^2}$, CoV $= 0.30$
Analysis B — mix composition			
Cement content	c	Normal	$\mu = 377,\ \sigma = 15\ \mathrm{kg\,m^{-3}}$
Aggregate content	agg	Normal	$\mu = 1920,\ \sigma = 50\ \mathrm{kg\,m^{-3}}$
Water/cement ratio	w/c	Uniform	$[0.30,\ 0.38]$
Silica/cement ratio	s/c	Uniform	$[0.08,\ 0.12]$

5.3.3 Quantity of interest and limit state

Both analyses share the same scalar quantity of interest (QoI): the *peak* gas pressure of the 10 mm history,

$$p_g^{10} \;=\; \max_t \, p_g(x = 10 \text{ mm}, \, t). \tag{5.1}$$

This near-surface gauge is the most critical location for thermo-hydral spalling, it was a calibration target in Chapter 4, and it is the one depth the model reproduces reliably (the deeper probes are not validated, Section 4.4.2), so it is a trustworthy and experimentally anchored output. Spalling is expressed by the limit-state function (Appendix B.5.1)

$$G(\mathbf{X}) = p_{\text{crit}} - p_g^{10}(\mathbf{X}), \tag{5.2}$$

where $\mathbf{X}$ is the input vector of the analysis considered (K_0 in A; $(c, agg, w/c, s/c)$ in B) and p_{crit} is the critical pressure tied to the temperature-degraded tensile strength. Failure corresponds to $G \leq 0$.

5.4 Uncertainty propagation with OpenTURNS

5.4.1 Coupling OpenTURNS to the Cast3M solver

Uncertainty quantification is carried out with the OpenTURNS library [**baudin2017**], reusing the Python automation layer built for the calibration sweeps of Section 4.1.2. There, `itertools.product` is also reused for a probabilistic design.

Besides, extra steps is required for the automated stochastic analyses:

1. Define the distribution(Table 5.1), using `ComposedDistribution`.
2. Generate an LHS design of N samples (Section 5.4.2).
3. Python script for post-process and ploting: correlation, Sobol indices, response PDF.

5.4.2 Latin Hypercube Sampling

Monte Carlo (Appendix B.2) converges only as $1/\sqrt{N}$ and needs many runs of an expensive THM model. Latin Hypercube Sampling (LHS) is used instead as a variance-reduced alternative [**mckay1979**, **baudin2017**]: each marginal is split into N equiprobable strata with one sample per stratum, so the input space is covered evenly and the output statistics stabilise with markedly fewer Cast3M evaluations. For the small number of variables here, the Gaussian-quadrature scheme of Appendix B.3 offers an even cheaper deterministic cross-check, used in Section 5.7.

5.5 Sensitivity analysis

5.5.1 Analysis A — permeability K_0 (A_Γ constant)

In Analysis A only K_0 is random; all other inputs, including the permeability–hydration coefficient A_Γ (`AK`), are held at their calibrated values. Fixing A_Γ is deliberate: as shown in Section 4.3.1, K_0 controls the *magnitude* of the peak whereas A_Γ mainly shifts its *timing*, and the two must be balanced to keep the Newton–Raphson solver stable. Keeping A_Γ constant isolates the magnitude-driving uncertainty and avoids the divergence reported in Chapter 4.

With a single random variable the analysis simply evaluates how the gas pressure p_g^{10} respond to K_0 over its lognormal distribution (mean $K_0^{\text{cal}} = 8.5 \times 10^{-20}$ m^2, CoV $= 0.30$); the relationship is strongly negative: lower permeability traps more vapour, which results in a higher pressure peak. A Sobol decomposition is not meaningful here, as a single variable trivially accounts for 100% of the variance $S_1 = S_T = 1$.

Therefore, for Analysis A, OpenTURNS focuses two main sensitivity outputs (Figures 5.1 and 5.2): the input–output correlation, and the bootstrap confidence-interval bands of the indices.

5.5.2 Analysis B — mix composition

In Analysis B, the four mix proportions $(c, agg, w/c, s/c)$ are treated as random variables while the intrinsic permeability K_0 is held fixed. Because evaluating variance-based sensitivity measures directly from the computationally expensive Cast3M model via a limited 30-sample LHS design is mathematically insufficient and highly inaccurate, a surrogate modeling strategy is necessarily employed.

To overcome this computational bottleneck, a surrogate model is first constructed within OpenTURNS using Polynomial Chaos Expansion (PCE). This metamodel maps the input mix proportions to the peak gas pressure p_g^{10}, effectively approximating the complex non-linear THM finite element solver into a fast analytical function based on the LHS training data.

Once the PCE metamodel is built and validated, it allows for a highly accurate and computationally inexpensive extraction of the sensitivity measures. Consequently, this analysis produces four main outputs (Figures 5.3 to 5.6): the input–output correlation

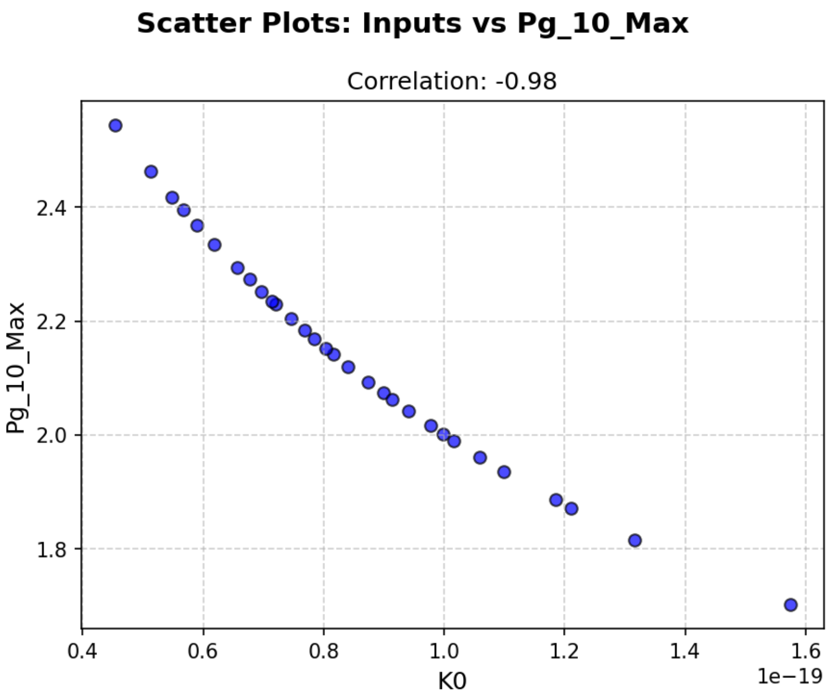


Figure 5.1: Analysis A— correlation: Pearson and Spearman rank correlation of K_0 with the peak pressure p_g^{10}.

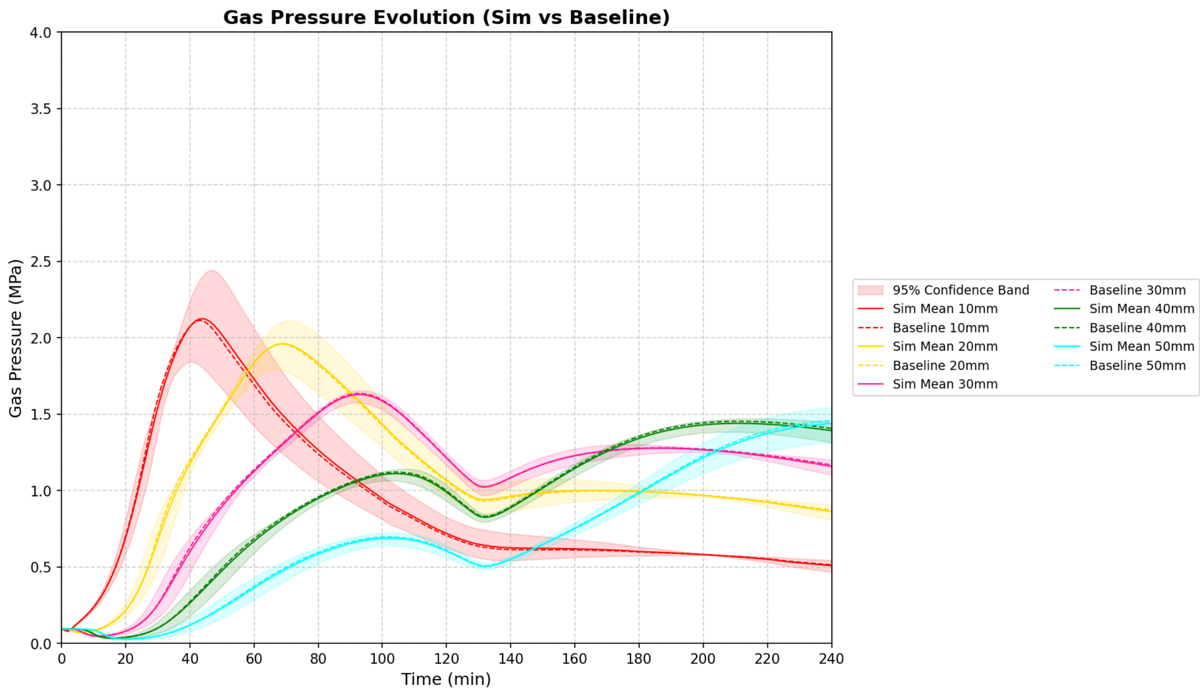


Figure 5.2: Analysis A — bootstrap 95% confidence-interval bands of the first- and total-order Sobol indices.

(evaluated from the raw LHS samples), the first- and total-order Sobol' indices (derived directly from the PCE coefficients), the second-order Sobol' indices (cross-interactions evaluated via massive sampling on the metamodel), and the bootstrap confidence-interval bands of the indices.

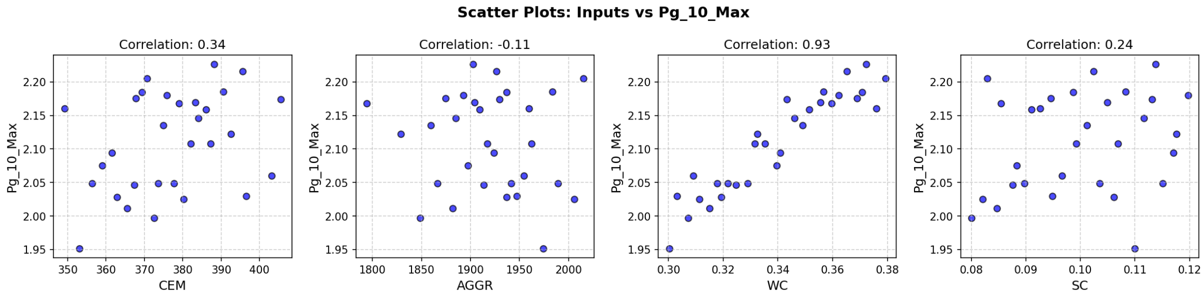


Figure 5.3: Analysis B — correlation: Pearson and Spearman rank correlation of each mix proportion $(c, agg, w/c, s/c)$ with the peak pressure p_g^{10}.

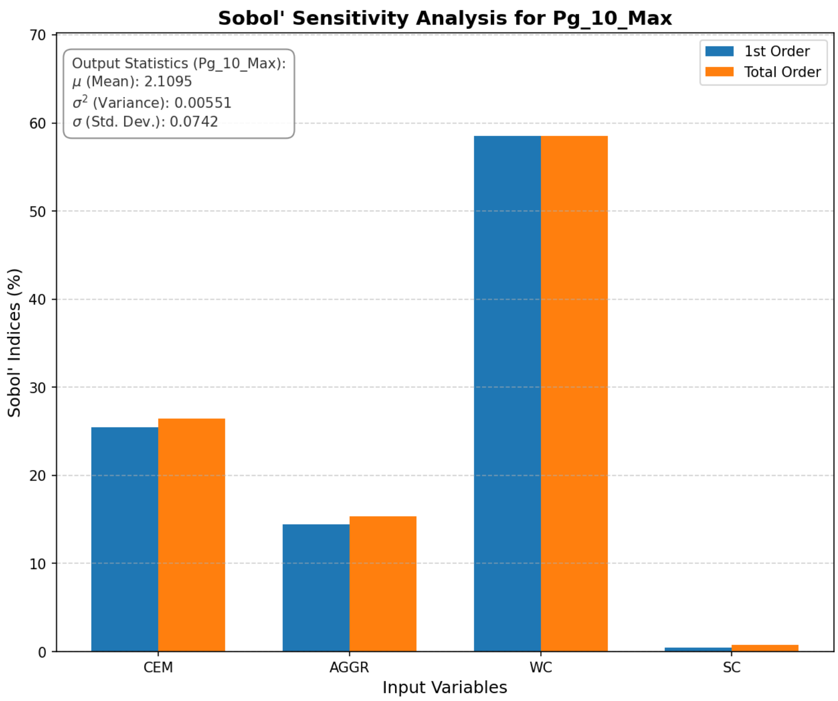


Figure 5.4: Analysis B — first- and total-order Sobol indices (S_i, S_{T_i}) of the mix proportions on p_g^{10}.

[REVIEW]

Analysis. Among the mix proportions, w/c is expected to dominate — it sets the porosity and the initial saturation, i.e. both the ease of vapour escape and the size of the water reservoir — followed by s/c, which densifies the paste, while the cement and aggregate *contents* (c, agg) should contribute little. Non-negligible second-order indices for the pair $(w/c, s/c)$ would indicate that the two ratios jointly shape the pore structure. Any proportion with a negligible total index can be fixed at its mean

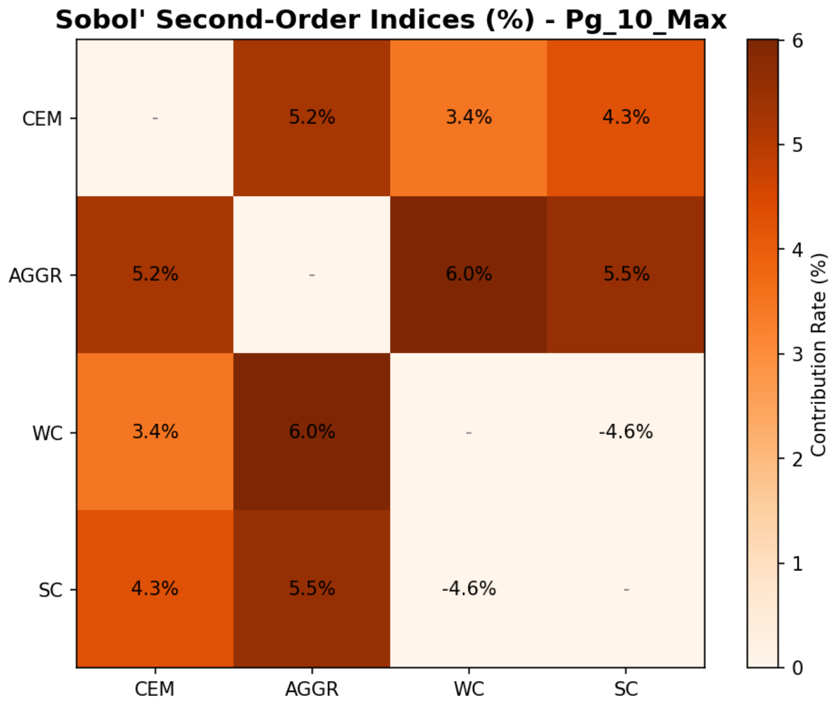


Figure 5.5: Analysis B — second-order Sobol indices S_{ij} of the mix-proportion pairs on p_g^{10}.

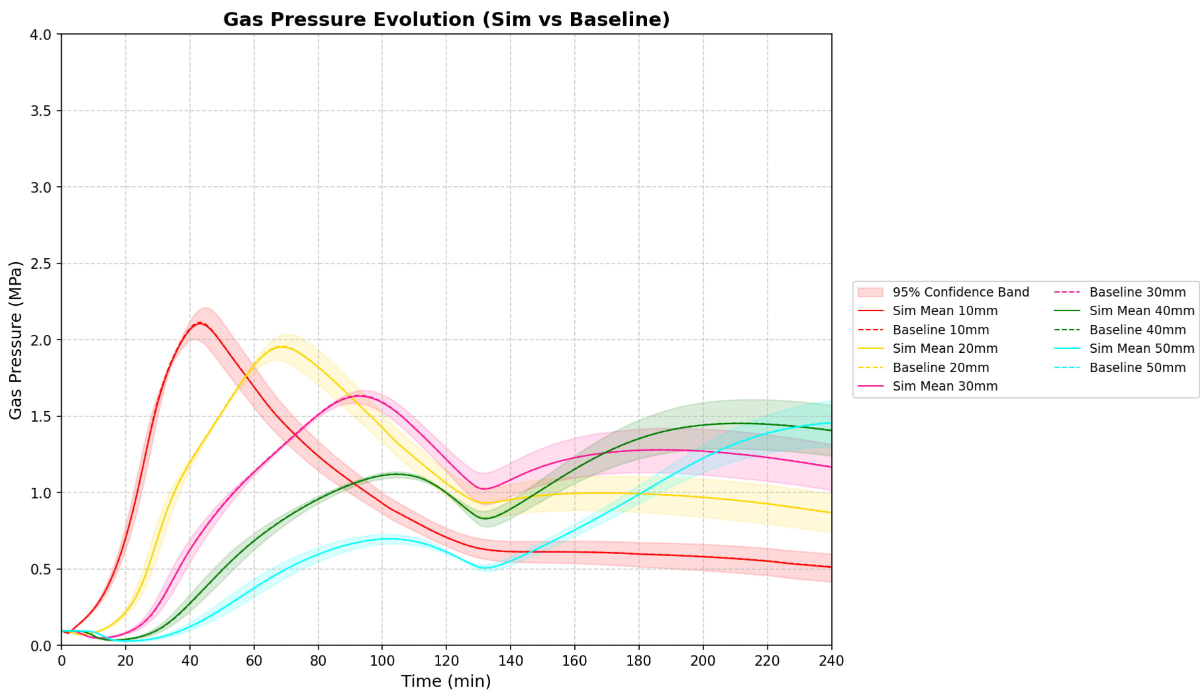


Figure 5.6: Analysis B — bootstrap 95% confidence-interval bands of the first- and total-order Sobol indices.

in the reliability step (Section 5.6).

5.6 Reliability analysis

For each analysis, the propagated sample yields a discrete set of structural responses (e.g., the peak gas pressure p_g^{10} or the longitudinal shrinkage strain). To turn this input scatter into a quantitative safety statement, the continuous Probability Density Function (PDF) and Cumulative Distribution Function (CDF) must be reconstructed. With the limit state of Equation (B.26) and the formulas of Appendix B.5, the probability of failure is calculated as the upper-tail area $P_f = 1 - F(p_{\text{crit}})$ and the reliability index as $\beta = -\Phi^{-1}(P_f)$.

Depending on the stochastic method employed, two distinct numerical strategies are utilized to reconstruct the distributions and evaluate the failure probability:

Kernel Density Estimation (Analyses A, C1, and C2): For analyses where the input uncertainty is limited to a single random variable (e.g., global K_0) or governed by a spatial random field (`ALEA`), building a multidimensional metamodel is either mathematically unnecessary or algorithmically incompatible. Instead, the continuous PDF is reconstructed directly from the raw, macroscopic Cast3M outputs using non-parametric Kernel Density Estimation (KDE). The probability of failure is then directly estimated by integrating the tail of this smoothed empirical distribution.

Monte Carlo on PCE Metamodel (Analysis B): In the multivariable mix-composition study, evaluating the failure probability directly from the limited 30-sample LHS design would yield highly inaccurate tail estimations. Instead, a massive Monte Carlo simulation (10^5 realizations) is executed exclusively on the Polynomial Chaos Expansion (PCE) surrogate model—which was previously constructed for the sensitivity analysis (Section 5.5.2). This allows for a highly accurate and computationally inexpensive evaluation of P_f without requiring additional Cast3M runs.

These reconstructed PDFs (Figures 5.7 and 5.8) visually demonstrate how the input uncertainties propagate towards the critical threshold.

[REVIEW]

Analysis. Comparing the two distributions (Figures 5.7 and 5.8) shows which source of uncertainty contributes more to the exceedance of p_{crit}: the permeability scatter of Analysis A is expected to produce the wider spread of p_g^{10} and hence the larger P_f (smaller β), the mix-composition uncertainty of Analysis B a narrower one. Placing the Chapter 4 calibrated value and the measured Kalifa 10 mm peak on each

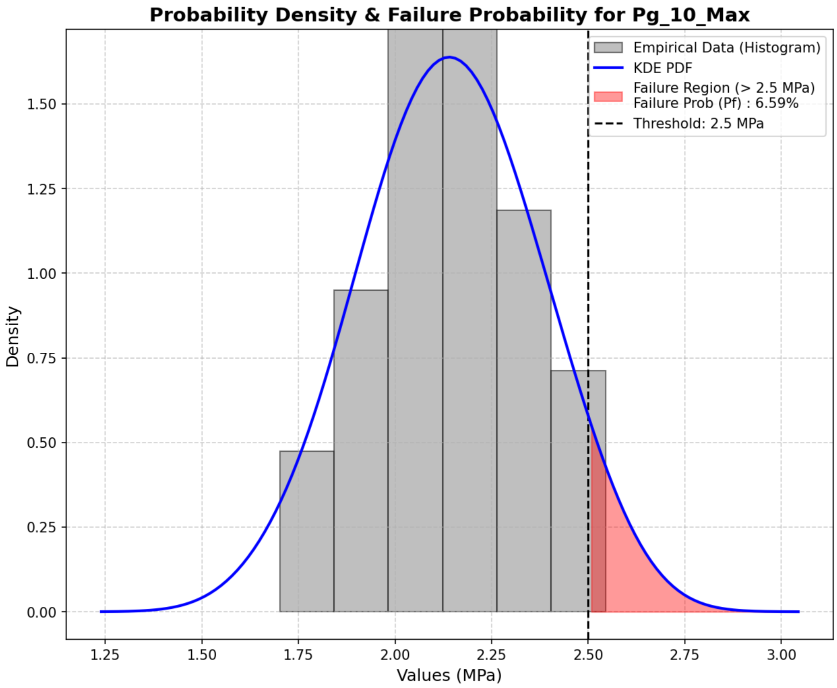


Figure 5.7: Analysis A: PDF of the peak pressure p_g^{10} under the permeability uncertainty (K_0), with the critical threshold p_{crit} (shaded tail $= P_f$) and the measured Kalifa 10 mm peak.

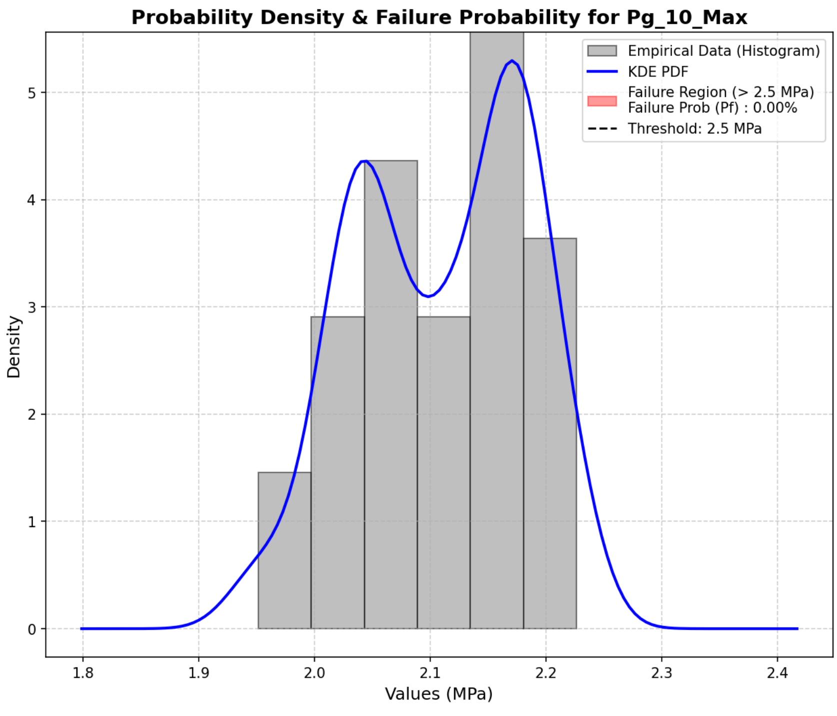


Figure 5.8: Analysis B: PDF of the peak pressure p_g^{10} under the mix-composition uncertainty $(c, agg, w/c, s/c)$, with the critical threshold p_{crit} (shaded tail $= P_f$) and the measured Kalifa 10 mm peak.

axis anchors the estimate experimentally. The value of p_{crit} retained (tied to the temperature-degraded tensile strength) and the resulting P_f and β for each analysis are reported here.

5.7 Effect of spatial heterogeneity on structural safety

The analyses above treat the permeability as a single random variable, constant over the specimen in a given realisation. Real concrete varies from point to point, and local weak zones — not average properties — govern failure. The intrinsic permeability K_{int}, the most spatially variable and most influential property (Section 5.5.1), is therefore additionally described as a *random field* and compared with the homogeneous description. The same lognormal marginal as in Analysis A is used (mean $= K_0^{\text{cal}} = 8.5 \times 10^{-20}$ m^2, CoV $= 0.30$, calibrated in Chapter 4).

5.7.1 A simplified isothermal poro-mechanical model

Propagating a spatially heterogeneous random field through the fully coupled, highly non-linear Thermo-Hygro-Chemical (THC) model presents severe computational and algorithmic challenges. Because the thermal and hydric properties evolve dynamically with temperature and hydration, assembling a parametric system matrix with a *fixed* structure — a strict requirement for the Gaussian-quadrature framework (`QUADRATU`/`PARASTAT`, Appendix B.3) in Cast3M — is mathematically intractable. Furthermore, executing numerous non-linear THM realizations is computationally prohibitive and highly prone to divergence.

To obtain a controlled, tractable, and stable comparison of the stochastic methods, a *simplified isothermal poro-mechanical model* applied to a $10 \times 10 \times 10$ cm concrete cube is adopted instead (simulated exploiting its symmetries). By holding the temperature constant at room conditions ($T = 20°$C), the complex thermo-chemical constitutive evolution is entirely frozen. Consequently, the intrinsic permeability is treated in a classical macroscopic manner, decoupled from the effective hydration degree (i.e., the parameter A_Γ of the THC formulation is deactivated). The quantity entering the assembled flow matrix is strictly the liquid-phase effective permeability, which becomes a function of the capillary pressure P_c only:

$$K_{\text{eff}}(P_c) \;=\; \frac{k_{rl}(P_c)\,\rho_l\,K_0}{\mu_l}, \tag{5.3}$$

where k_{rl} is the liquid relative permeability governed by the van Genuchten model, ρ_l is the liquid density, and μ_l is the dynamic viscosity.

In this test case, the specimen is subjected to convective drying, where the internal relative humidity drops from an initial state of 90% to an environmental humidity of 50% at the exposed surfaces. The intrinsic permeability K_0 (`KINT`) is designated as the spatially variable input. The primary outputs of interest are the local capillary pressure P_c and the resulting *longitudinal shrinkage strain*. This strain is induced purely by capillary forces through the poro-mechanical coupling, where the skeletal pressure acting on the solid matrix is explicitly computed as $p_s = -S_l(P_c)P_c$. While this simplified approach sacrifices the high-temperature thermal coupling and the hydration-dependent permeability, it perfectly isolates the effect of K_0 spatial heterogeneity on the hydro-mechanical response in a stable setting where all three stochastic methods can be applied consistently.

5.7.2 Stochastic propagation methods

The uncertainty of the intrinsic permeability (K_0) is propagated using three distinct stochastic approaches, all detailed in Appendix B:

Analysis C1 (OpenTURNS + LHS): A homogeneous probabilistic baseline where K_0 is treated as a random variable uniform over the entire specimen (i.e., one single value per realization). A Latin Hypercube Sampling (LHS) design of 30 samples is generated via OpenTURNS (Section 5.4.2).

Analysis C2 (Cast3M `ALEA`): A heterogeneous random field approach. The spatially variable field $K_0(\mathbf{x})$ is generated directly on the 3D finite element mesh using the Turning Bands Method (TBM) coupled with a lognormal transformation (Appendices B.2.3 and B.2.4). The spatial correlation length is set to $L_c = 0.02$ m. A total of 30 independent runs are executed to capture the spatial variability effects.

Analysis C3 (Gaussian Quadrature): A deterministic, few-point integration scheme utilizing the `QUADRATU` operator in Cast3M (Appendices B.3 and B.3.4). K_0 is modeled homogeneously using 7 quadrature points. Within Cast3M, the statistical moments of the response are computed via the `PARASTAT` operator, and the corresponding continuous PDF can be theoretically approximated as a Pearson type III distribution using the `PRODENS` operator (detailed in Appendix B).

However, for Analysis C3, because Cast3M lacks native support for exporting these continuous PDF curves for high-quality visualization, statical moments calculation and the Pearson type III reconstruction based on the quadrature moments was re-implemented in Python. This ensures consistent plotting and comparison against the LHS and random field methods. Ultimately, this approach serves as a computationally inexpensive cross-check to validate the statistical moments obtained from the LHS baseline (C1).

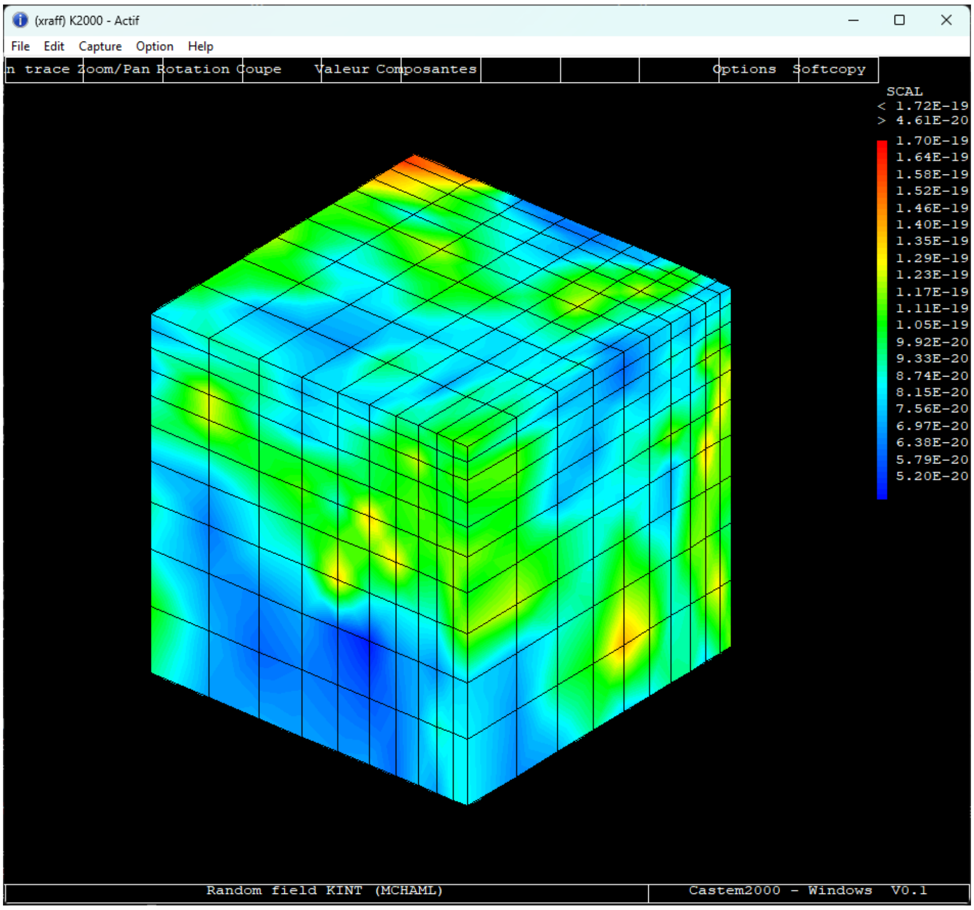


Figure 5.9: A single realization of the spatially heterogeneous intrinsic permeability field $K_0(\mathbf{x})$ generated on the 3D cube mesh using the Turning Bands Method ($L_c = 0.02$ m).

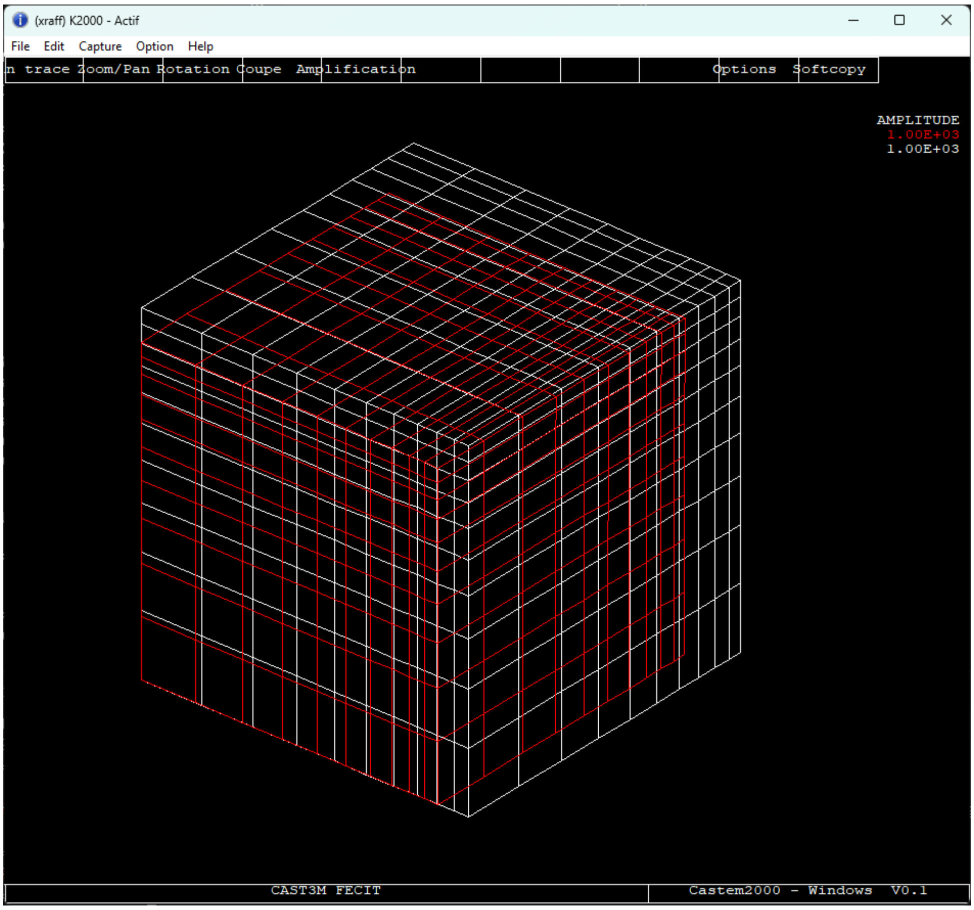


Figure 5.10: Resulting final shrinkage strain field of the concrete cube induced by capillary forces, highlighting the localized deformations driven by the heterogeneous permeability field.

Figure 5.11: Mean longitudinal shrinkage strain history at the corner point for the two stochastic analyses, LHS (C1) and the spatial random field (C2). Faint lines show the individual realisations of each analysis.

Figure 5.11 compares the mean shrinkage history of the two analyses before each is examined in detail. The ALEA mean (C2) lies slightly below the LHS mean (C1) throughout the drying period, while its realisations are visibly less scattered. These two features pull in opposite directions: a lower mean points to a more severe average response, but a tighter spread limits how often large strains are reached. The mean curve alone therefore cannot identify the more critical analysis. This requires comparing the full distributions of the final strain, given by the probability density functions in **??** and **??**.

Conclusion. Two cross-checks and one primary physical observation are drawn from this comparative study. First, while both LHS (Analysis C1) and Gaussian quadrature (Analysis C3) treat the intrinsic permeability as a homogeneous random variable, their resulting probability distributions exhibit noticeable differences. Although the 7-point quadrature scheme provides a computationally inexpensive estimation, it yields a *more*

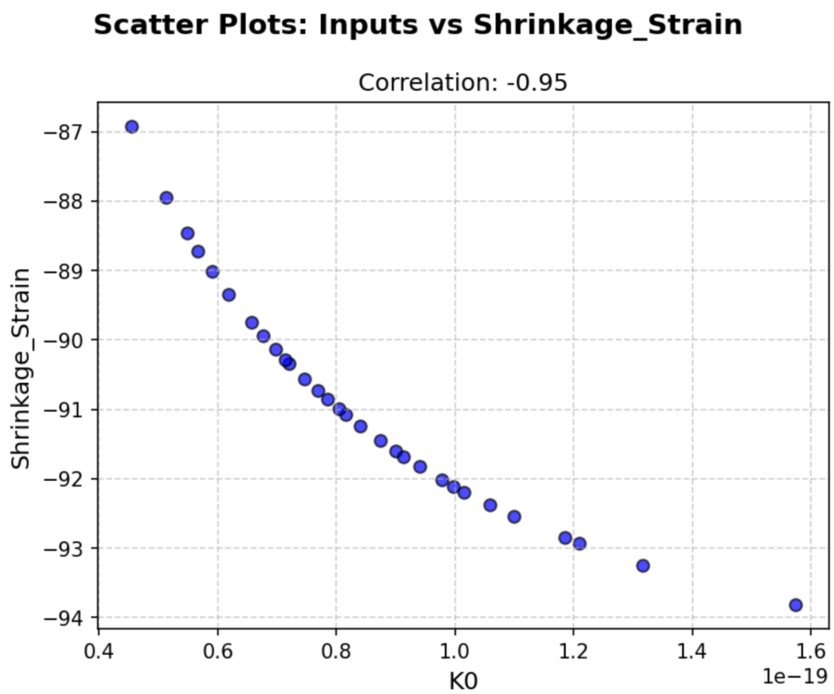


Figure 5.12: Analysis C1 (OpenTURNS + LHS): Pearson and Spearman rank correlation between the intrinsic permeability (K_0) and the resulting longitudinal shrinkage strain.

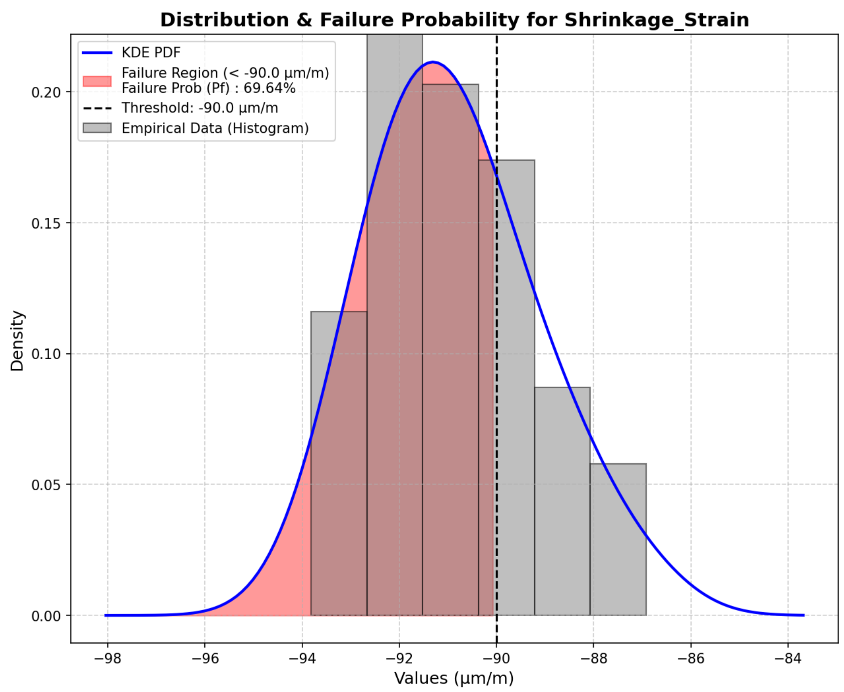


Figure 5.13: Analysis C1 (OpenTURNS + LHS): Probability Density Function (PDF) of the longitudinal shrinkage strain induced by the uncertainty of the intrinsic permeability (K_0).

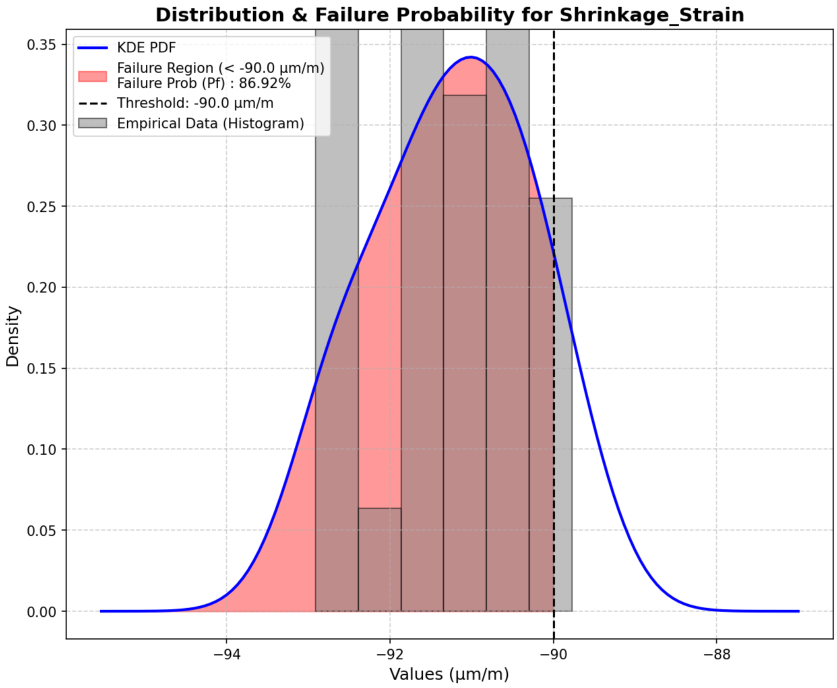


Figure 5.14: Analysis C2 (Cast3M `ALEA`): Probability Density Function (PDF) of the longitudinal shrinkage strain considering the spatial heterogeneity of the intrinsic permeability field.

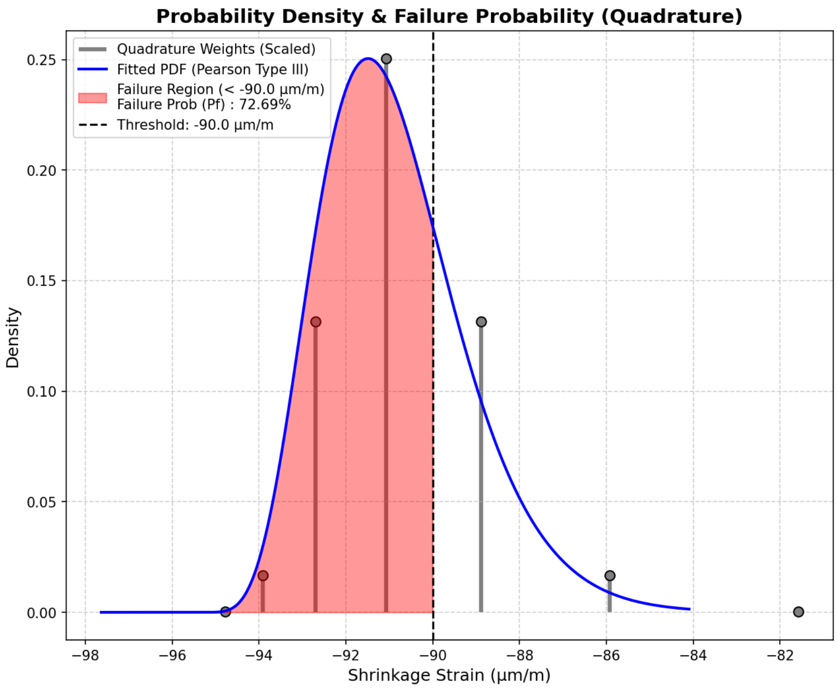


Figure 5.15: Analysis C3 (Gaussian Quadrature): Probability Density Function (PDF) of the longitudinal shrinkage strain evaluated through the deterministic few-point integration scheme.

critical response profile for the capillary pressure P_c and the longitudinal shrinkage strain compared to the 30-sample LHS baseline.

Second, and most importantly, the spatially heterogeneous random-field approach (Analysis C2) produces the *most critical* structural response. Physically, local low-permeability zones restrict moisture transport and amplify local capillary forces, which severely concentrates the deformation. This mechanism generates significantly higher peak shrinkage strains and a wider probability tail than both homogeneous cases (C1 and C3), because the highly stressed local weak zones cannot be entirely compensated by the surrounding material.

The 30-realisation ensemble used for the Monte Carlo/LHS and random field analyses is a pragmatic estimate given the computational cost of each non-linear run; its scatter indicates the residual statistical uncertainty. Ultimately, the significant gap between the homogeneous and heterogeneous results quantitatively highlights how a simple random-variable assumption dangerously underestimates the structural risk. The full Turning Bands / `ALEA` construction, the quadrature operators, and the Monte Carlo convergence details are collected in Appendix B.

Chapter 6

Conclusions and Perspectives

[REVIEW] This chapter is under review and should be limited to 2 pages

6.1 Main findings

(1) THM model with hydration-degree implemented and runnable in Cast3M;

(2) temperature calibrated well, matching Kalifa and Dauti;

(3) gas pressure not calibrated accurately regardless of parameter adjustment –> raises questions about the nature of the measured quantity;

(4) sensitivity: indicates the dominant parameter group (via first and second order Sobol);

(5) reliability: quantifies the risk of spalling from the 10 mm gas pressure layer near the surface;

(6) spatial heterogeneity (random field) gives unfavorable results compared to the homogeneity assumption.

6.2 Model limitations and possible improvements

The TH–>M model is one-dimensional (staggered, creep removed);

the random-field ALEA cannot yet be fully integrated into the THM model due to computational cost; gas pressure calibration is still open.

6.3 Perspectives for future research

Notable points include:

(a) redesigning/re-proposing gas pressure measurements to verify actual measured values;

(b) full coupling of two-dimensional M–>H (damage→permeability);

(c) integrating random field directly into THM via surrogate/metamodel to reduce Monte Carlo costs;

(d) extending THC to full service life (curing–>aging–>accident) as in Sciumè 2024;

(e) verification on different concrete types/heating scenarios.

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Appendix A

Thermo-Hydro-Mechanical behavior of Heated Concrete

[REVIEW] This chapter is under review

A.1 Thermal behavior

A.1.1 Conduction, convection and radiation

Heat transfer is the transport of thermal energy driven by a temperature difference. The following definitions of heat transfer modes are summarized from standard references [3].

Conduction

Conduction is the transfer of energy from the more energetic to the less energetic particles of a substance

- **Physical Mechanism:**
- **Rate Equation (Fourier's Law):** For a one-dimensional plane wall having a temperature distribution $T(x)$, the rate equation is expressed as Fourier's law:

$$q''_{cond} = -\lambda \frac{dT}{dx} \tag{A.1}$$

  Where q''_{cond} is the heat flux (W/m^2), λ is the thermal conductivity $(W/m \cdot K)$, and the minus sign indicates that heat is transferred in the direction of decreasing temperature.

Convection

The convection heat transfer mode is

- **Physical Mechanism:** Convection heat transfer occurs between a fluid in motion and a bounding surface when the two are at different temperatures
- **Rate Equation (Newton's Law of Cooling):** Regardless of the nature of the convection process, the appropriate rate equation is:

$$q''_{conv} = h(T_s - T_\infty) \tag{A.2}$$

  Where q'' is the convective heat flux (W/m^2), h is the convection heat transfer coefficient ($W/m^2 \cdot K$), T_s is the surface temperature, and T_∞ is the fluid temperature.

Radiation

Thermal radiation is energy emitted by matter that is at a nonzero temperature, and unlike conduction and convection, it does not require the presence of a material medium.

- **Physical Mechanism:** The energy of the radiation field is transported by electromagnetic waves (or photons)
- **Emissive Power (Stefan-Boltzmann law):** The upper limit to the emissive power (rate of energy released per unit area) for an ideal radiator or blackbody is given:

$$E_b = \sigma T_s^4 \tag{A.3}$$

  The heat flux emitted by a real surface is given by $E = \varepsilon\sigma T_s^4$, where σ is the Stefan-Boltzmann constant and ε is the emissivity.
- **Net Radiation Exchange:** For a gray surface at T_s completely surrounded by a much larger isothermal surface at T_{sur}, the net rate of radiation heat transfer per unit area is:

$$q''_{rad} = \varepsilon\sigma(T_s^4 - T_{sur}^4) \tag{A.4}$$

A.1.2 Heat transfer

Heat Equation

Based on the principle of energy conservation and Fourier's law, the general heat diffusion equation (or heat equation) for a medium is detailed in [7] and can be expressed as:

$$\rho c_p \frac{\partial T}{\partial t} + \mathrm{div}(-\lambda \overrightarrow{\mathrm{grad}}(T)) - q = 0 \tag{A.5}$$

Where:

- ρ is the volumetric mass density (kg/m^3).
- c_p is the specific heat capacity ($J/kg \cdot K$).
- λ is the thermal conductivity ($W/m \cdot K$).
- T is the temperature field and t is time.
- q is the volume heat flux source (W/m^3).

To solve this equation, appropriate boundary conditions must be specified:

- **Prescribed temperatures :** The temperature is specified as $T = T_{imp}$ on the boundary surface ∂V^T.
- **Prescribed surface heat flux :** The total heat flux due to conduction, convection, and radiation on the boundary surface ∂V^φ is given by:

$$\vec{n} \cdot (\lambda \overrightarrow{\mathrm{grad}}(T)) = \varphi_{imp} + h(T_f - T) + \varepsilon\sigma(T_\infty^4 - T^4) \tag{A.6}$$

  where φ_{imp} is the prescribed heat flux, $h(T_f - T)$ represents the convection term, and $\varepsilon\sigma(T_\infty^4 - T^4)$ is the radiation term.

Thermal properties

The thermal properties of concrete are strongly influenced by both the hydric state and the material composition.

The overall heat capacity ρC_p reflects the multiphase nature of the porous medium, combining contributions from the solid skeleton, liquid water, dry air, and water vapor [**lewis1998**]:

$$\rho C_p = (1-\phi)\,\rho_s\,C_{ps} + \phi\,[S_l\,\rho_l\,C_{pl} + (1-S_l)\,(\rho_a\,C_{pa} + \rho_v\,C_{pv})] \tag{A.7}$$

where:

The solid density decreases linearly with temperature due to dehydration: $\rho_s(T) = \rho_{s0}[1 - A_\rho(T - T_0)]$ with $\rho_{s0} = 2611$ kg/m^3.

The liquid water density evolves with temperature according to a polynomial fit [**furbish1997**]:

$$\rho_l(T) = \sum_{i=0}^{5} a_i T^i \tag{A.8}$$

with coefficients a_i given in Table **??**.

[REVIEW] This section is under review for matching the THC model implemented. Cps is not updated The solid thermal capacity evolves linearly with temperature [**harmathy1973**]:

$$C_{ps}(T) = C_{ps0}[1 + A_c(T - T_0)] \tag{A.9}$$

with $C_{ps0} = 940$ J·kg^{-1}K^{-1}.

The thermal conductivity depends on saturation, porosity, and temperature [**baggio1993**]:

$$\lambda = \lambda_d \left(1 + 4\,\frac{S_l\,\phi\,\rho_l}{(1-\phi)\,\rho_s}\right) \tag{A.10}$$

with [**harmathy1973**]:

$$\lambda_d = \lambda_{d0}\,[1 + A_\lambda\,(T - T_0)] \tag{A.11}$$

As drying progresses ($S_l \to 0$), both ρC_p and λ decrease, which accelerates the penetration of the thermal front into the material. These dependencies highlight the influence of the hydric state on the thermal field, which is further discussed in Appendix A.4.1.

Discretization

Using the FE discretization, the temperature and temperature gradient in the spatial domain are expressed in matrix form as:

$$T(x) = [N(x)]\{T\}, \quad \overrightarrow{\text{grad}}(T) = [B(x)]\{T\} \tag{A.12}$$

From the weak formulation of the heat equation combined with the spatial discretization, we obtain the general matrix equation:

$$[C]\{\dot{T}\} + [K]\{T\} = \{F\} \tag{A.13}$$

Where the matrices are defined as follows:

- **Capacity matrix:**

$$[C] = \int_V \rho c_p [N]^T [N] dV \quad (J/K) \tag{A.14}$$

- **Conductivity matrix:**

$$[K] = \int_V [B]^T[\lambda][B]dV + \int_{\partial V\varphi} h[N]^T[N]dS \quad (W/K) \tag{A.15}$$

- **Equivalent nodal flux vectors:**

$$\{F\} = \int_V [N]^T q dV + \int_{\partial V\varphi} [N]^T \left(\varphi_{imp} + hT_f + \varepsilon\sigma(T_\infty^4 - T^4)\right) dS \quad (W) \tag{A.16}$$

For steady-state thermal analysis, the temperature is time-independent ($\{\dot{T}\} = 0$) (assuming constant material properties and neglecting radiation):

$$[K]\{T\} = \{F\} \tag{A.17}$$

By solving this linear system, we can determine the unknown temperatures $\{T\}$ at the nodes across the entire global mesh.

A.1.3 Thermal gradient

Due to the low thermal conductivity of concrete, fire exposure produces steep temperature gradients near the heated surface. The hot surface layer tends to expand while the cooler interior restrains this deformation, generating compressive stresses at the surface and tensile stresses in the interior.

Since concrete is inherently weak in tension, this stress state can initiate cracking parallel to the heated face. The role of thermal gradient in the spalling process is discussed further in Appendix A.5.2.

A.2 Hydric behavior

[REVIEW] This section is under review for matching the THC model implemented

[REVIEW]

When concrete is exposed to high temperatures, the behavior of water within the porous network plays a central role in the thermo-hydro-mechanical response of the material. This section presents the classification of water in concrete, the phase-change mechanisms, the governing equation for moisture transport, and the constitutive laws required to close the system.

A.2.1 Types of water in concrete

Concrete is a multiphase porous material whose voids can be filled with water, air and other fluids. The water present in the cement paste can be broadly classified into two categories [5]

- **Pore water (free/evaporable water):** This water evaporates when the thermodynamic conditions are met (*i.e.*, at $T = 100\,^{\circ}\mathrm{C}$ and atmospheric pressure) and exists in three main forms:
  - *Capillary and inter-hydrate water:* fills the capillary pores (size $\sim$ 100s of nm) and inter-hydrate pores(size $\sim$ 10s of nm), corresponding to a saturation $S_{ssp} < S_l < 1$.
  - *Physically adsorbed water (gel water):* retained by the surface of the solid matrix due to Van der Waals forces, confined in gel pores (size $\sim$ a few nm).
  - *Interlayer water:* confined in $\sim$ 1 nm space between the C-S-H layers.
- **Chemically bound water (non-evaporable):** This water is combined with anhydrous cement during hydration and forms part of the crystalline structure of hydration products (e.g., C-S-H, portlandite). It can only be released through thermal decomposition (dehydration) at elevated temperatures.

It is important to note that the range of pore sizes is a continuous one; the dimensional boundaries between capillary, inter-hydrate and gel pores are approximations to inherently blurred boundaries.

A.2.2 Evaporation and dehydration

Two distinct phase-change mechanisms govern the release of water in heated concrete:

- **Evaporation** is the transition of free liquid water into vapor. It occurs progressively as temperature increases, becoming significant above approximately $100\,^{\circ}\mathrm{C}$ at atmospheric pressure. The evaporation rate $\dot{m}_{vap}$ is determined from the mass conservation of liquid water (see Appendix A.2.3).
- **Dehydration** is the thermally-driven decomposition of hydration products, releasing chemically bound water as vapor. The main reactions occur at characteristic temperature ranges [5]:
  - Up to $300\,^{\circ}\mathrm{C}$: progressive decomposition of C-S-H phases and ettringite;
  - 400–500 °C: dehydroxylation of portlandite ($Ca(OH)_2 \rightarrow CaO + H_2O$);

– $\sim 570\,°\mathrm{C}$: α–β quartz transformation;

– Above $600\,°\mathrm{C}$: decarbonation of calcareous aggregates ($CaCO_3 \rightarrow CaO + CO_2$).

Both evaporation and dehydration act as vapor sources that increase gas pressure within the pore network. They also consume latent energy, affecting the temperature field (see Appendix A.4).

Dehydration kinetics

The cumulative dehydrated water mass is described by the following phenomenological relationship [**pesavento2000**]:

$$\Delta m_{dehyd} = f_s \cdot f_m \cdot f_c \cdot f(T) \tag{A.18}$$

where f_s is a stoichiometric factor, f_m accounts for the concrete's age, f_c is the cement mass per unit volume, and $f(T)$ is a temperature-dependent function:

$$f(T) = \begin{cases} 0 & \text{if } T < 105\,°\mathrm{C} \\ \left[1 + \sin\!\left(\frac{\pi}{2}\,(1 - 2\exp(-0.004\,(T - 105)))\right)\right] & \text{if } T \geq 105\,°\mathrm{C} \end{cases} \tag{A.19}$$

Enthalpy of evaporation

The enthalpy of evaporation is temperature-dependent and is approximated by the Watson formula [**reid1987**]:

$$\Delta H_{vap}(T) = \Delta H_{vap,0} \left(\frac{T_{crit} - T}{T_{crit} - T_0}\right)^{0.38} \tag{A.20}$$

where $T_{crit} = 647$ K is the critical temperature of water and $\Delta H_{vap,0} = 2.672 \times 10^5$ J/kg.

A.2.3 Moisture transport equation

Transport mechanisms

The transport of moisture in heated concrete is governed by two mechanisms:

- **Advective flow (Darcy's law):** The bulk flow of liquid water and gas through the porous network is driven by pressure gradients:

$$\mathbf{v}_{l-s} = -\frac{K\,k_{rl}}{\mu_l}\,(\nabla p_g - \nabla p_c) \tag{A.21}$$

$$\mathbf{v}_{g-s} = -\frac{K\, k_{rg}}{\mu_g} \nabla p_g \tag{A.22}$$

where K is the intrinsic permeability, k_{rl} and k_{rg} are the relative permeabilities of liquid and gas, μ_l and μ_g are the dynamic viscosities, and $p_l = p_g - p_c$ is the liquid pressure.

- **Diffusive flow (Fick's law):** Water vapor diffuses within the gas mixture according to:

$$\mathbf{j}_v = -\rho_g\, D\, \nabla \left(\frac{\rho_v}{\rho_g} \right) \tag{A.23}$$

where D is the effective diffusivity accounting for the tortuous porous network.

Mass conservation equations

The mathematical framework for moisture transport in heated concrete is based on the conservation of mass for each phase. The general form of a mass conservation equation for a phase π is [5]:

$$\frac{\partial}{\partial t}(\text{density}) + \nabla \cdot (\text{flux}) = \text{source} \tag{A.24}$$

Concrete at high temperature is treated as a multiphase porous medium consisting of a solid skeleton (s), liquid water (l), water vapor (v), and dry air (a). Applying Equation (A.24) to each phase yields [5]:

- **Solid skeleton:**

$$\frac{\partial m_s}{\partial t} = \dot{m}_{dehyd} \tag{A.25}$$

- **Liquid water:**

$$\frac{\partial m_l}{\partial t} + \nabla \cdot (m_l\, \mathbf{v}_{l-s}) = -\dot{m}_{vap} - \dot{m}_{dehyd} \tag{A.26}$$

- **Water vapor:**

$$\frac{\partial m_v}{\partial t} + \nabla \cdot (m_v\, \mathbf{v}_{g-s}) + \nabla \cdot (m_v\, \mathbf{v}_{v-g}) = \dot{m}_{vap} \tag{A.27}$$

- **Dry air:**

$$\frac{\partial m_a}{\partial t} + \nabla \cdot (m_a\, \mathbf{v}_{g-s}) + \nabla \cdot (m_a\, \mathbf{v}_{a-g}) = 0 \tag{A.28}$$

where $\mathbf{v}_{\pi-s}$ is the velocity of phase π ($\pi = l, g$) relative to the solid skeleton, and $\mathbf{v}_{\pi-g}$ is the velocity of species π ($\pi = v, a$) relative to the gas mixture. The source term $\dot{m}_{vap}$ represents the evaporation rate ($\dot{m}_{vap} > 0$ for evaporation, < 0 for condensation), and $\dot{m}_{dehyd}$ is the dehydration rate ($\dot{m}_{dehyd} < 0$, treated as a mass loss of the solid skeleton and a mass source for liquid water).

The mass of each phase per unit volume of porous medium is [5]:

$$m_s = (1-\phi)\,\rho_s \tag{A.29}$$
$$m_l = \rho_l\, S_l\, \phi \tag{A.30}$$
$$m_v = \rho_v\,(1-S_l)\,\phi \tag{A.31}$$
$$m_a = \rho_a\,(1-S_l)\,\phi \tag{A.32}$$

where ϕ is the porosity and S_l is the liquid saturation degree.

By combining the mass conservation of liquid water and water vapor, the evaporation source term $\dot{m}_{vap}$ is eliminated, yielding the **water species conservation equation**:

$$\begin{aligned} &S_l\rho_l\frac{\partial \phi}{\partial t} + S_l\phi\frac{\partial \rho_l}{\partial t} + \rho_l\phi\frac{\partial S_l}{\partial t} + (1-S_l)\phi\frac{\partial \rho_v}{\partial t} - \rho_v\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_v\frac{\partial \phi}{\partial t} \\ &+\nabla\cdot\left(-K\frac{\rho_l k_{rl}}{\mu_l}\nabla p_l\right) + \nabla\cdot\left(-K\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_vM_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd} \end{aligned} \tag{A.33}$$

This equation has three primary unknowns: the temperature T, the capillary pressure p_c, and the gas pressure p_g. Similarly, the **dry air conservation equation** reads:

$$(1-S_l)\phi\frac{\partial \rho_a}{\partial t} - \rho_a\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_a\frac{\partial \phi}{\partial t} + \nabla\cdot\left(-K\frac{\rho_a k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_vM_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0 \tag{A.34}$$

Constitutive relations for the gas phase

The gaseous phase (dry air + water vapor) is governed by the following relations [5]:

- **Ideal gas law:**

$$\rho_\pi = \frac{p_\pi M_\pi}{RT}, \quad \pi = a, v \tag{A.35}$$

  where M_π is the molar mass and R is the universal gas constant.

- **Dalton's law:**

$$p_g = p_a + p_v \tag{A.36}$$

- **Kelvin equation** (thermodynamic equilibrium between liquid and vapor):

$$p_v = p_{vs}(T)\,\exp\left(\frac{M_v}{\rho_l RT}(p_g - p_c - p_{vs})\right) \tag{A.37}$$

  where $p_{vs}(T)$ is the saturation vapor pressure, approximated by the Hyland-Wexler relationship [**ashrae2001**]:

$$p_{vs}(T) = \exp\left[\frac{C_1}{T} + C_2 + C_3T + C_4T^2 + C_5T^3 + C_6\ln(T)\right] \tag{A.38}$$

Sorption isotherms

The relationship between capillary pressure and liquid saturation is described by the Van Genuchten model [**vangenuchten1980**, **baroghel1999**]:

$$p_c = a\left(S_l^{-b} - 1\right)^{1-1/b} \tag{A.39}$$

where a and b are material parameters determined experimentally. Representative values at ambient temperature are given in Table A.1.

Table A.1: Van Genuchten parameters at ambient temperature

	NSC	HPC	NSM	HPM
a [MPa]	18.62	46.94	37.55	96.28
b [-]	2.275	2.060	2.168	1.954

At elevated temperatures, the parameter a is modified to account for changes in surface tension and pore structure [**pesavento2000**]:

$$S_l = \left(\left(\frac{E(T)}{a(T)}\,p_c\right)^{b/(b-1)} + 1\right)^{-1/b} \tag{A.40}$$

where $E(T)$ accounts for the surface tension evolution and $a(T)$ accounts for microstructure changes above 100 °C.

Intrinsic permeability

Permeability is a key parameter governing moisture transport. For a thermo-hydral problem, the following relationship is adopted [**gawin1999**]:

$$K = K_0 \cdot 10^{A_T(T-T_0)}\left(\frac{p_g}{p_{atm}}\right)^{A_p} \tag{A.41}$$

where K_0 is the intrinsic permeability at reference conditions, $A_T = 0.005$ and $A_p = 0.368$.

When mechanical damage is considered, the permeability increases significantly [**gawin2002**]:

$$K = K_0 \cdot 10^{A_T(T-T_0)}\left(\frac{p_g}{p_{atm}}\right)^{A_p} \cdot 10^{A_D\,\mathcal{D}} \tag{A.42}$$

where $\mathcal{D}$ is the damage parameter and A_D is a material constant. This damage-permeability coupling is a key mechanism in the THM model (see Appendix A.4).

Relative permeability

The relative permeabilities of liquid and gas are described by the Van Genuchten relationships [**vangenuchten1980**]:

$$k_{rl} = \sqrt{S_l}\left(1 - \left(1 - S_l^{1/A_S}\right)^{A_S}\right)^2 \tag{A.43}$$

$$k_{rg} = \sqrt{1 - S_l}\left(1 - S_l^{1/A_S}\right)^{2A_S} \tag{A.44}$$

where $A_S = 0.6$.

Effective diffusivity

The diffusivity of vapor in air at a given temperature and pressure is [**perre1990**]:

$$D_{va}(T, p_g) = D_{v0}\left(\frac{T}{T_0}\right)^{B_v}\frac{p_{g0}}{p_g} \tag{A.45}$$

where $D_{v0} = 2.58 \times 10^{-5}$ m^2/s and $B_v = 1.667$. The effective diffusivity in the porous medium accounts for saturation and tortuosity:

$$D = (1 - S_l)^{A_v}\,\phi\,\tau\,D_{va}(T, p_g) \tag{A.46}$$

with the tortuosity:

$$\tau(\phi, S_l) = \phi^{1/3}(1 - S_l)^{7/3} \tag{A.47}$$

Fluid viscosities

The dynamic viscosities are temperature-dependent [**ashrae2001**]. The gas viscosity is a concentration-weighted average:

$$\mu_g = \frac{p_v}{p_g}\mu_v + \frac{p_a}{p_g}\mu_a \tag{A.48}$$

where μ_v and μ_a are linear functions of temperature. The liquid water viscosity decreases strongly with temperature [**thomas1995**]:

$$\mu_l = \mu_{l0}\exp\left(\frac{-\alpha_l(T - T_0)}{T}\right) \tag{A.49}$$

with $\mu_{l0} = 1.002 \times 10^{-3}$ Pa·s and $\alpha_l = 0.6612$.

Porosity evolution

The porosity increases with temperature due to microstructural changes [**schneider1989**]:

$$\phi = \phi_0 + A_\phi (T - T_0) \tag{A.50}$$

where ϕ_0 is the initial porosity and A_ϕ is a material constant.

A.2.4 Pore pressure build-up

When concrete is heated, evaporation and dehydration produce vapor that raises the pore pressure. Part of this vapor migrates toward the cooler interior, where it recondenses and forms a quasi-saturated zone — the *moisture clog*. This layer blocks further gas transport, causing significant pressure build-up behind the drying front. In dense concretes such as HPC, the low intrinsic permeability K_0 further limits gas evacuation.

When the pore pressure exceeds the tensile strength, it can trigger explosive spalling (Appendix A.5.2).

A.3 Mechanical behavior

[REVIEW]

The mechanical behaviour is modelled by coupling linear elasticity with continuum damage mechanics (Mazars model), described in the following subsections.
The TH and mechanical problems are solved with a staggered (sequential) solution strategy: the TH subsystem is solved first to obtain p_g, p_c, T, which are then passed as known loading to the mechanical step.
The specific coupling method between the TH and mechanical solvers has not yet been finalized and will be detailed and modified in Chapter 2 and Chapter 3 as the study progresses.

A.3.1 Linear elastic model

In the present work, concrete is modeled as a linear elastic material. The effective stress tensor $\tilde{\boldsymbol{\sigma}}$, defined as the stress acting on the undamaged material, is related to the elastic strain tensor $\boldsymbol{\varepsilon}_e$ through the isotropic Hooke's law:

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}_e \tag{A.51}$$

where **C** is the constant fourth-order elastic stiffness tensor, defined by the Young's modulus E and Poisson's ratio ν.

The total strain is decomposed into mechanical and free (stress-free) components, as detailed in Appendix A.4.2. Since creep is neglected in this work, the elastic strain reduces to:

$$\boldsymbol{\varepsilon}_e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh} \tag{A.52}$$

which enters directly into Hooke's law (Equation (A.51)).

Linear finite element formulation

The spatial discretization of the momentum conservation equation (Equation (A.83)) is carried out using the standard Galerkin finite element method [4].

The continuous displacement field $\mathbf{u}$ is approximated by nodal displacements $\{\mathbf{U}\}$ using shape functions $\mathbf{N}$, and the strain is obtained via the compatibility matrix $\mathbf{B}$ such that $\boldsymbol{\varepsilon} = \mathbf{B}\{\mathbf{U}\}$.

Applying the principle of virtual work, the global system takes the form:

$$[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{A.53}$$

where $[\mathbf{K}_M] = \int_V \mathbf{B}^T \mathbb{C}\, \mathbf{B}\, dV$ is the elastic stiffness matrix. Its generalisation to account for damage is given in Equation (A.63).

The load vector $\{\mathbf{F}\}$ receives contributions from external forces and from the thermo-hydric fields: (Appendix A.4.1):

$$\{\mathbf{F}\} = \{\mathbf{f}_{ext}\} + \underbrace{\int_V \mathbf{B}^T \mathbb{C}\, \boldsymbol{\varepsilon}^{th}\, dV + \int_V \mathbf{B}^T \mathbb{C}\, \boldsymbol{\varepsilon}^{sh}\, dV}_{\text{strain decomposition (Equation (A.71))}} + \underbrace{\int_V \mathbf{B}^T\, \alpha\, p_s\, \mathbf{I}\, dV}_{\text{effective stress (Equation (A.77))}} \tag{A.54}$$

where $\{\mathbf{f}_{ext}\}$ groups the classical external forces. All thermo-hydric terms are computed from the TH solution at each time step and enter as prescribed loading, detailed in Appendix A.4.1.

In Cast3M, this system is assembled using the `RIGI` and `RESO` operators for the linear part. When damage is included, the non-linear iterative solution is handled by `PASAPAS` (Equation (A.64)).

A.3.2 Mazars damage model

Concrete cracks when tensile strains become too large. In this work, this behaviour is represented with the Mazars damage model [**mazars1986**, 9].

Damage parameter and equivalent strain

The model introduces a single scalar damage variable $\mathcal{D} \in [0, 1]$ [**mazars1986**]. The effective stress in the damaged material is:

$$\boldsymbol{\sigma}' = (1 - \mathcal{D})\, \mathbb{C}\, \boldsymbol{\varepsilon}_{el} \tag{A.55}$$

When $\mathcal{D} = 0$, this reduces to Hooke's law (Equation (A.51)). The damage variable is driven by the equivalent strain, computed from the positive principal elastic strains:

[REVIEW]

Because concrete cracking is primarily governed by tensile extensions, Mazars introduced an *equivalent strain* (ε_{eq}) to translate any 3D multiaxial strain state into a scalar uniaxial indicator. In the case of a thermo-mechanical loading, it is important to note that only the elastic strain tensor $\boldsymbol{\varepsilon}_e$ contributes to the evolution of damage. The equivalent strain is calculated solely from the positive principal elastic strains ε_i:

$$\varepsilon_{eq} = \sqrt{\langle \boldsymbol{\varepsilon}_e \rangle_+ : \langle \boldsymbol{\varepsilon}_e \rangle_+} = \sqrt{\sum_{i=1}^{3} \langle \varepsilon_i \rangle_+^2} \tag{A.56}$$

where the Macaulay brackets $\langle x \rangle_+$ denote the positive part of x (i.e., $\langle x \rangle_+ = x$ if $x \geq 0$, and 0 if $x < 0$).

Damage initiates when the equivalent strain exceeds the initial damage threshold, denoted as ε_{d0}.

$$\varepsilon_{d0} = \frac{f_t}{E}. \tag{A.57}$$

At the damage onset ($\varepsilon_{eq} = \varepsilon_{d0}$), the damage variable is zero:

$$\mathcal{D} = 0 \qquad \text{for} \qquad \varepsilon_{eq} \leq \varepsilon_{d0}. \tag{A.58}$$

For uniaxial tension, damage onset corresponds to:

$$\sigma_t = f_t. \tag{A.59}$$

In this simplified interpretation, f_t denotes the uniaxial tensile strength of the concrete and E is Young's modulus.

Damage evolution laws for tension and compression

To accurately represent concrete behavior, the total damage D is formulated as a combination of two independent damage variables: D_t for tension and D_c for compression.

When the damage threshold is exceeded ($\varepsilon_{eq} > \varepsilon_{d0}$), the evolution of these two modes is explicitly governed by exponential laws:

$$\mathcal{D}_t = 1 - \frac{(1 - A_t)\varepsilon_{d0}}{\varepsilon_{eq}} - A_t \exp\left[-B_t(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{A.60}$$

$$\mathcal{D}_c = 1 - \frac{(1 - A_c)\varepsilon_{d0}}{\varepsilon_{eq}} - A_c \exp\left[-B_c(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{A.61}$$

where A_t, B_t, A_c, and B_c are material parameters identified from uniaxial tensile and compressive tests. These parameters modulate the shape of the post-peak softening curves: parameters A define the residual asymptotic stress level, while parameters B control the slope of the stress drop.

The global macroscopic damage $\mathcal{D}$ is then calculated using a linear combination with weighting coefficients α_t and α_c:

$$\mathcal{D} = \alpha_t^\beta \mathcal{D}_t + \alpha_c^\beta \mathcal{D}_c \tag{A.62}$$

The coefficients α_t and α_c depend on the principal stress state, evaluating the respective contributions of tensile and compressive stresses. In pure tension, $\alpha_t = 1$ and $\alpha_c = 0$; conversely, in pure compression, $\alpha_c = 1$ and $\alpha_t = 0$. The parameter β (typically fixed at 1.06) is a shear-correction coefficient introduced to improve the structural response under shear loadings.

Non-linear equilibrium and iterative solution

In the general case with damage, the stiffness matrix reads:

$$[\mathbf{K}_M] = \int_V \mathbf{B}^T (1 - \mathcal{D})\, \mathbb{C}\, \mathbf{B}\, dV \tag{A.63}$$

When $\mathcal{D} = 0$ (intact material), this reduces to the standard linear elastic stiffness matrix $\int_V \mathbf{B}^T \mathbb{C}\, \mathbf{B}\, dV$.

Due to damage evolution, the constitutive relation becomes non-linear and Equation (A.53) cannot be solved in a single step. The equilibrium is enforced iteratively using a Newton-Raphson-type algorithm [4]. At each iteration i, the stiffness matrix (Equation (A.63)) is evaluated with the current damage $\mathcal{D}^i$, and the correction is obtained from:

$$[\mathbf{K}_M]^i\{\delta\mathbf{U}\}^{i+1} = \{\mathbf{F}\} - \{\mathbf{R}\}^i \tag{A.64}$$

where:

- $\{\mathbf{R}\}^i = \int_V \mathbf{B}^T\boldsymbol{\sigma}^i\,dV$ is the internal force vector, with $\boldsymbol{\sigma}^i = (1-\mathcal{D}^i)\,\mathbb{C}\,\boldsymbol{\varepsilon}^i_{el}$;
- $\{\mathbf{F}\}$ is the external load vector including all thermo-hydric contributions (Equation (A.54)).

The displacements and damage are updated until convergence: $\|\{\mathbf{F}\} - \{\mathbf{R}\}^i\| < \epsilon\,F_{ref}$. In Cast3M, this iterative procedure is handled by the `PASAPAS` operator.

A.3.3 Mesh dependency and fracture energy regularization

A well-known drawback of local softening models such as Mazars' is the pathological mesh dependency: softening causes strain localization in a band whose width depends on the element size, leading to vanishing energy dissipation upon mesh refinement [**bazant1976**].

Two main families of regularization exist in the literature:

- **Nonlocal averaging:** the local equivalent strain ε_{eq} is replaced by a weighted spatial average $\bar{\varepsilon}_{eq}$ over a characteristic volume controlled by an internal length l_c [**pijaudier1987**, **giry2011**]. This is the approach adopted by [5].
- **Fracture energy regularization (Hillerborg):** the softening law parameters are adjusted element-by-element so that each element dissipates the same fracture energy G_f per unit crack area, regardless of its size [**hillerborg1976**]. This is the approach adopted in the present work, as it is simpler to implement in Cast3M and does not require additional nonlocal operators.

Principle of fracture energy regularization

Physically, the fracture energy G_f [**hillerborg1976**] represents the energy per unit area required to propagate a crack. In the continuum damage framework, cracking is smeared over the element volume. A characteristic length L_e bridges the two descriptions: in Cast3M (using `@LONGEF`), $L_e = V^{1/D}$ where D is the spatial dimension.

The total dissipated energy per unit crack area is:

$$G_f = G_{elastic} + G_{softening} \tag{A.65}$$

Elastic energy ($G_{elastic}$)

The elastic energy density up to damage onset is:

$$g_{elastic} = \frac{1}{2} f_t \, \varepsilon_{d0} = \frac{f_t^2}{2E_0} \tag{A.66}$$

so that $G_{elastic} = g_{elastic} \, L_e = \frac{f_t^2}{2E_0} \, L_e$, and the energy available for softening is:

$$G_{softening} = G_f - \frac{f_t^2}{2E_0} \, L_e \tag{A.67}$$

Regularization of B_t

Assuming $A_t \approx 1$ (brittle tension), the post-peak stress simplifies to $\sigma \approx f_t \exp[-B_t(\varepsilon - \varepsilon_{d0})]$, whose integral gives $g_{softening} = f_t/B_t$. Equating $G_{softening} = g_{softening} \cdot L_e$ yields the modified parameter:

$$B_t^{mod} = \frac{f_t \, L_e}{G_f - \frac{f_t^2}{2E_0} \, L_e} \tag{A.68}$$

This ensures that every element dissipates exactly G_f regardless of mesh size, making the global response objective with respect to spatial discretization.

A.4 Coupling mechanisms

The thermal, hydric, and mechanical behaviors presented in the previous sections are not independent: they interact through a network of coupling mechanisms. This section identifies and formulates the two main coupling paths that govern the response of concrete exposed to fire.

A.4.1 Thermo-Hydric (TH) coupling mechanisms

The thermal and hydric fields are coupled in both directions: temperature drives phase changes and modifies transport properties, while fluid flow and phase changes affect the energy balance.

Temperature as a driver of hydric phenomena

Temperature enters the hydric conservation equations (Equations (A.33) and (A.34)) through two pathways:

- **Mass source terms:** evaporation and dehydration (Appendix A.2.2) are thermally activated processes that inject vapor into the pore network, appearing as source terms $\dot{m}_{vap}$ and $\dot{m}_{dehyd}$ in the mass balance equations;
- **Transport coefficients:** most constitutive parameters are temperature-dependent — permeability $K(T)$, porosity $\phi(T)$, fluid viscosities $\mu(T)$, saturating vapor pressure $p_{vs}(T)$, and diffusivity $D(T)$ — so the coefficients of the mass conservation equations evolve as the thermal field changes.

Energy conservation equation

The energy conservation equation governs the temperature field within the porous medium. Similar to the heat equation in Appendix A.1.2, it includes a storage term, a conductive flux, and additionally accounts for advective heat transport and phase-change energy terms [5]:

$$\rho C_p \frac{\partial T}{\partial t} + \underbrace{(m_l C_l \mathbf{v}_{l-s} + m_g C_g \mathbf{v}_{g-s}) \cdot \nabla T}_{\text{advection}} + \nabla \cdot (-\lambda \cdot \nabla T) = \underbrace{-H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd}}_{\text{phase-change energy}} \tag{A.69}$$

Compared to the classical heat equation Equation (A.5) $\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (\lambda \nabla T) = 0$, two additional contributions appear:

- **Advective heat transport:** The bulk flow of liquid water and gas, driven by pressure gradients, transport enthalpy through the material. This term redistributes energy along the flow paths and can be significant when permeability and pressure gradients are large.
- **Phase-change energy (right-hand side):** Concrete does not self-generate heat; these terms represent energy consumed by evaporation ($-H_{vap}\,\dot{m}_{vap}$, heat sink since $\dot{m}_{vap} > 0$) and dehydration ($+H_{dehyd}\,\dot{m}_{dehyd}$, also a heat sink since $\dot{m}_{dehyd} < 0$). Both locally slow the temperature rise.

After replacing the mass fluxes by Darcy's law (Equations (A.21) and (A.22)), the energy conservation takes its final coupled form:

$$\begin{aligned} \rho C_p \frac{\partial T}{\partial t} - K \left(C_l \frac{\rho_l k_{rl}}{\mu_l} (\nabla p_g - \nabla p_c) + C_g \frac{\rho_g k_{rg}}{\mu_g} \nabla p_g \right) \cdot \nabla T \\ - \nabla \cdot (\lambda \cdot \nabla T) = -H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd} \end{aligned} \tag{A.70}$$

where the overall heat capacity ρC_p (Equation (A.7)) and thermal conductivity λ (Equation (A.10)) are defined in Appendix A.1.2.

Thus, the thermal and hydric fields are *strongly coupled*: temperature modifies the source terms and transport coefficients in the moisture equation, while moisture content and fluid flow modify the heat capacity, conductivity, and introduce latent heat and advection terms in the energy equation.

A.4.2 Thermo-hydric input to the mechanical field

The mechanical behaviour of heated concrete is driven by the thermal and hydric fields. The coupling is *one-way*: the converged TH fields (T, p_g, p_c) serve as prescribed loading to the mechanical problem at each time step. The feedback from mechanics to TH through damage-enhanced permeability is discussed in Appendix A.5.1.

The TH fields enter the mechanical problem through two mechanisms: free strains and pore pressure loading, detailed in the following subsections.

Strain decomposition

When a mechanically loaded concrete element is exposed to high temperature, the total strain is decomposed as a superposition of components of different origins [5]:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_\sigma + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{sh} \tag{A.71}$$

where:

- $\boldsymbol{\varepsilon}_\sigma$ represents the *mechanical strains* induced by external loading (elastic strain, creep strain, etc.);
- $\boldsymbol{\varepsilon}_{th}$ represents the *thermal dilation strain*, driven by the temperature field;
- $\boldsymbol{\varepsilon}_{sh}$ represents the *capillary shrinkage strain*, driven by the capillary pressure field.

In general, the mechanical strain includes elastic and creep contributions:

$$\boldsymbol{\varepsilon}_\sigma = \boldsymbol{\varepsilon}_{el} + \boldsymbol{\varepsilon}_{creep} \tag{A.72}$$

Since creep is neglected in this work, $\boldsymbol{\varepsilon}_{creep} = \mathbf{0}$ and thus $\boldsymbol{\varepsilon}_\sigma = \boldsymbol{\varepsilon}_{el}$

Thermal dilation

The thermal dilation strain is isotropic and produces a uniform expansion of the material, which is expressed in incremental form as [5]:

$$d\boldsymbol{\varepsilon}_{th} = \beta_s(T)\, dT\, \mathbf{I} \tag{A.73}$$

where $\mathbf{I}$ is the second-order identity tensor and $\beta_s(T)$ is the thermal dilation coefficient, which for concrete can be approximated as a linear function of temperature:

$$\beta_s(T) = A_{B1}\,(T - T_0) + A_{B0} \tag{A.74}$$

with A_{B0} and A_{B1} being material constants.

Capillary shrinkage

The drying process modifies the pore pressure state, which in turn induces a volumetric deformation of the solid skeleton. The capillary shrinkage strain is expressed as [5]:

$$d\boldsymbol{\varepsilon}_{sh} = \frac{\alpha}{K_T}\,\frac{dp_s}{dt}\,\mathbf{I} \tag{A.75}$$

where α is the Biot coefficient, K_T is the bulk modulus of the drained skeleton, and p_s is the average pore pressure acting on the solid skeleton, defined as:

$$p_s = p_g - S_l\, p_c = S_g\, p_g + S_l\, p_l \tag{A.76}$$

This expression represents the saturation-weighted average of the gas and liquid pressures. Since p_g and p_c are direct outputs of the TH problem, p_s is entirely determined by the TH solution.

Effective stress and pore pressure loading

For a porous medium saturated by multiple fluids, the total stress is decomposed into the part carried by the solid skeleton and the part carried by the pore fluids. Following [**lewis1998**, 5], the effective stress is defined as:

$$\boldsymbol{\sigma}' = \boldsymbol{\sigma}^{Total} + \alpha\, p_s\, \mathbf{I} \tag{A.77}$$

where $\boldsymbol{\sigma}^{Total}$ is the total stress tensor, α is Biot's coefficient, and p_s is the average pore pressure (Equation (A.76)).

The effective stress $\boldsymbol{\sigma}'$ is the stress that governs the mechanical behaviour of the skeleton. In the finite element implementation, this relation is not applied directly; instead, the pore pressure term $\alpha\, p_s$ enters the load vector as an external loading (Equation (A.54)), and the effective stress is recovered from the elastic strain after solving for displacements:

$$\boldsymbol{\sigma}' = (1 - \mathcal{D})\, \mathbb{C}\, \boldsymbol{\varepsilon}_{el} \tag{A.78}$$

where $\boldsymbol{\varepsilon}_{el} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh}$ (Equation (A.52)). This effective stress is the input to the Mazars damage criterion (**??**). In heated concrete, the build-up of gas pressure (Appendix A.2.4) increases p_s in the load vector, which shifts $\boldsymbol{\sigma}'$ toward tension and promotes damage when f_t is exceeded.

A.5 THM model and spalling

This section assembles the governing equations of the thermo-hydro-mechanical (THM) model from the conservation laws and constitutive relations developed in the previous sections. The coupled system is then used to explain how spalling arises from the interaction between thermal, hydric, and mechanical fields.

A.5.1 Full THM governing formulation

The state of concrete at high temperature is described by four primary variables [5]:

$$\mathcal{U} = \{p_g,\ p_c,\ T,\ \mathbf{u}\} \tag{A.79}$$

where p_g is the gas pressure, p_c is the capillary pressure, T is the temperature, and $\mathbf{u}$ is the displacement vector.

The model consists of four conservation equations, each associated with one primary variable. Their final forms, after substitution of Darcy's law, Fick's law, and the constitutive relations, are as follows.

Dry air conservation

$$\begin{aligned} &\phi(1 - S_l)\left(\frac{\partial \rho_a}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial \rho_a}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial \rho_a}{\partial p_g}\frac{\partial p_g}{\partial t}\right) + (1 - S_l)\rho_a \frac{\partial \phi}{\partial T}\frac{\partial T}{\partial t} \\ &- \phi\rho_a\left(\frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t}\right) - \nabla\cdot\left(\frac{K\rho_a k_{rg}}{\mu_g}\nabla p_g\right) - \nabla\cdot\left(D\rho_g \frac{M_v M_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0 \end{aligned} \tag{A.80}$$

Water species conservation

$$
\begin{aligned}
&(S_l\rho_l - (1-S_l)\rho_v)\frac{\partial \phi}{\partial T}\frac{\partial T}{\partial t} + S_l\phi\frac{\partial \rho_l}{\partial T}\frac{\partial T}{\partial t} + \phi(\rho_l-\rho_v)\left(\frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t}\right)\\
&+(1-S_l)\phi\left(\frac{\partial \rho_v}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial \rho_v}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial \rho_v}{\partial p_g}\frac{\partial p_g}{\partial t}\right)\\
&-\nabla\cdot\left(\frac{K\rho_l k_{rl}}{\mu_l}(\nabla p_g - \nabla p_c)\right) - \nabla\cdot\left(\frac{K\rho_v k_{rg}}{\mu_g}\nabla p_g\right) - \nabla\cdot\left(D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd}
\end{aligned}
\tag{A.81}
$$

Energy conservation

$$
\begin{aligned}
\rho C_p\frac{\partial T}{\partial t} - K\left(C_l\frac{\rho_l k_{rl}}{\mu_l}(\nabla p_g - \nabla p_c) + C_g\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g\right)\cdot\nabla T - \nabla\cdot(\lambda\cdot\nabla T)\\
= -H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd}
\end{aligned}
\tag{A.82}
$$

Momentum conservation

By neglecting inertial forces:

$$\nabla\cdot\boldsymbol{\sigma}^{Total} + \rho\,\mathbf{g} = \mathbf{0} \tag{A.83}$$

where $\mathbf{g}$ is the gravitational acceleration and the total density is:

$$\rho = (1-\phi)\rho_s + \phi\,(S_l\rho_l + (1-S_l)\rho_g) \tag{A.84}$$

The total stress $\boldsymbol{\sigma}^{Total}$ is related to the effective stress via Bishop's relation (Equation (A.77)), which in turn depends on the primary variables p_g and p_c.

The discretisation of this equation into the global system $[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\}$ is presented in Appendix A.3.

Remark: The governing equations presented in this chapter — dry air conservation, water species conservation, energy conservation, and momentum conservation — together with the constitutive laws, form a complete THM model for concrete exposed to fire. These equations are continuous partial differential equations defined on the domain Ω. Their numerical solution requires spatial and temporal discretization, which is the subject of the following Chapter 3.

A.5.2 Spalling mechanisms

Spalling is defined as the violent or non-violent detachment of layers or pieces of concrete from the surface of a structural element during or after exposure to rapidly rising temperatures. It may lead to premature structural failure through reinforcement exposure and cross-section reduction. Several catastrophic tunnel fires (Channel Tunnel 1996, Mont-Blanc 1999, St. Gotthard 2001) have illustrated the severity of this phenomenon [5].

Despite extensive experimental research, the mechanisms behind spalling are not yet fully understood and no general consensus on a single theory has been reached. The two principal mechanisms most broadly accepted are presented below, followed by their combined action.

Thermal gradient mechanism

[REVIEW] This section is currently under development and will be refined as the study progresses.

When one face of a concrete member is exposed to fire, a steep temperature gradient develops through the thickness. The resulting differential thermal dilation produces (see Figure **??**):

1. **Compressive stresses in the heated cover:** The surface layers tend to expand but are restrained by the cooler interior, generating significant compressive stresses parallel to the heated surface.
2. **Tensile stresses in the interior:** The restraint of the expanding surface layers induces tensile stresses in deeper layers of the material.
3. **Fracture and energy release:** When the tensile stress exceeds the tensile strength f_t, cracks develop parallel to the surface. The sudden release of elastic energy stored in the compressed surface layers can drive the explosive ejection of concrete fragments.

High-performance concrete (HPC) is particularly prone to this mechanism due to its higher brittleness: more elastic energy is stored before failure, leading to a more violent release.

Pore pressure mechanism (moisture clog)

[REVIEW] This section is currently under development and will be refined as the study progresses.

The sequence of events is (see Figure **??**):

1. **Formation of a dry zone:** As the surface is heated, free water evaporates and chemically bound water is released as vapor through dehydration. A dry zone forms near the heated face.

2. **Vapor migration:** Due to pressure gradients, part of the vapor is evacuated toward the heated face, while part is driven toward the cooler interior.

3. **Condensation and moisture clog:** The inward-migrating vapor condenses when it reaches regions where the thermodynamic conditions are satisfied ($T < T_{sat}$). This process continues until a quasi-saturated zone is formed, known as the *moisture clog*, which acts as an impermeable barrier preventing further vapor migration toward the cold side.

4. **Pressure build-up:** Behind the moisture clog, the gas pressure p_g increases sharply as continued evaporation feeds vapor into a region from which it cannot escape. In addition, the thermal dilation of liquid water in saturated pores also contributes to pressure build-up.

5. **Tensile failure:** When the pore pressure exceeds the tensile strength of the material, cracks develop parallel to the surface, potentially leading to explosive spalling.

The key material parameters governing this mechanism are the *intrinsic permeability* K_0 and the *initial moisture content*. Low permeability hinders vapor evacuation, while high moisture content provides more water to be evaporated, both promoting higher pore pressures.

Combined mechanism

[REVIEW] This section is currently under development and will be refined as the study progresses.

In practice, explosive spalling generally occurs under the *combined action* of pore pressures and thermal stresses. The spalling condition can be expressed schematically as:

$$\sigma_{thermal}(T) + \sigma_{pore}(p_g, p_c, S_l) \geq f_t(T) \tag{A.85}$$

where $\sigma_{thermal}$ represents the thermal stress from differential dilation, σ_{pore} represents the stress from pore pressure loading (via effective stress), and $f_t(T)$ is the temperature-degraded tensile strength. When the sum of stresses exceeds the tensile strength, cracks develop parallel to the heated surface, often leading to violent failure of the fire-exposed region.

This combined criterion highlights why the complete THM model is needed: neither a purely thermal analysis (T only) nor a purely hydral analysis (p_g, p_c only) can capture the full picture.

A.6 Discretization

A.6.1 Spatial discretization

[REVIEW]

The specific coupling method mentioned in Appendix A.3 between the TH and mechanical solvers has not yet been finalized and will be updated in Chapter 2 and Chapter 3 as the study progresses.

Applying the standard Galerkin finite element method to the three TH conservation equations (Equations (A.80) to (A.82)) yields the semi-discrete system [5]:

$$\mathbf{C}\,\dot{\mathbf{Y}} + \mathbf{K}\,\mathbf{Y} = \mathbf{f} \tag{A.86}$$

where $\mathbf{Y} = \{\bar{p}_g, \bar{p}_c, \bar{T}\}$ is the vector of nodal TH unknowns. The operators $\mathbf{C}$ (capacity), $\mathbf{K}$(conductivity), and $\mathbf{f}$ (forcing) are 3×3 block matrices:

$$\mathbf{K} = \begin{pmatrix} K_{gg} & K_{gc} & K_{gt} \\ K_{cg} & K_{cc} & K_{ct} \\ K_{tg} & K_{tc} & K_{tt} \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} C_{gg} & C_{gc} & C_{gt} \\ 0 & C_{cc} & C_{ct} \\ 0 & C_{tc} & C_{tt} \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} f_g \\ f_c \\ f_t \end{pmatrix} \tag{A.87}$$

The off-diagonal entries K_{ij}, C_{ij} $(i \neq j)$ encode the coupling between the thermal and hydric fields. Detailed expressions of each block are given in [5], Appendix A.

The mechanical problem (Equation (A.83)) is solved in a staggered manner: at each time step, the converged TH fields $\bar{T}$, $\bar{p}_g$, $\bar{p}_c$ enter as prescribed loading. The discrete equilibrium (Equation (A.53)) reads:

$$[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{A.88}$$

where $\mathbf{K}_M$ (Equation (A.63)) contains the damaged stiffness $(1-\mathcal{D})\,\mathbb{C}$ — the displacement-dependent part that remains on the left-hand side — and $\mathbf{F}$ (Equation (A.54)) collects the TH-prescribed loads ($\boldsymbol{\varepsilon}^{th}$, $\boldsymbol{\varepsilon}^{sh}$, $\alpha\,p_s$) that do not depend on $\mathbf{U}$.

A.6.2 Temporal discretization

[REVIEW] This section is currently under development and will be refined as the study progresses.

The time integration of Equation (A.86) is carried out using the Wilson-θ method [**lewis1998**, **zienkiewicz2013**], which evaluates the equation at an intermediate time $t_{n+\theta} = t_n + \theta\,\Delta t$ with $\theta \in [0, 1]$. This transforms the semi-discrete system into a fully discrete algebraic system at each time step:

$$\tilde{\mathbf{K}}\,\Delta\mathbf{Y}^{n+1} + \mathbf{K}\,\mathbf{Y}^{n} = \mathbf{f}^{n+\theta} \tag{A.89}$$

where:

$$\Delta\mathbf{Y}^{n+1} = \mathbf{Y}^{n+1} - \mathbf{Y}^{n}, \qquad \tilde{\mathbf{K}} = \frac{\mathbf{C}}{\Delta t} + \theta\,\mathbf{K} \tag{A.90}$$

The parameter θ controls the scheme: $\theta = 0$ gives the explicit forward Euler method, $\theta = 1$ the fully implicit backward Euler, and $\theta = 0.5$ the Crank-Nicolson scheme. In practice, $\theta = 1$ is used for unconditional stability.

Since the capacity and conductivity matrices depend nonlinearly on the unknowns ($\mathbf{C} = \mathbf{C}(\mathbf{Y})$, $\mathbf{K} = \mathbf{K}(\mathbf{Y})$), Equation (A.89) is solved iteratively using the Newton-Raphson method within each time step.

A.6.3 Solution strategy

[REVIEW] This section is currently under development and will be refined as the study progresses.

The TH subsystem (Equation (A.89)) is solved *monolithically*: all three unknowns p_g, p_c, T are updated simultaneously within a single Newton-Raphson loop at each time step.

The mechanical problem (Equation (A.88)) is solved in a *staggered* manner: at each time step, the converged TH solution provides $\bar{T}$, $\bar{p}_g$, $\bar{p}_c$, which are used to assemble $\mathbf{F}$ (Equation (A.54)) and solve for $\mathbf{U}$. The damage variable $\mathcal{D}$ computed from the Mazars model feeds back to the intrinsic permeability via Equation (A.42), creating a weak M→H coupling updated at the next time step.

Appendix B

Theory of Stochastic analysis

[REVIEW] This chapter is under review

B.1 Random variable and random field

B.1.1 Random variables (PDF, CDF, moments, CoV)

[REVIEW] This section is currently under development and will be refined as the study progresses.

A **random variable** X is a quantity whose value is not fixed but described by a probability law. Instead of saying "$E = 30$ GPa", one writes $E \sim \mathcal{N}(30, 3^2)$ GPa, meaning that the Young's modulus follows a normal (Gaussian) distribution with mean $\mu = 30$ GPa and standard deviation $\sigma = 3$ GPa.

PDF and CDF

The Probability Density Function (**PDF**) $f_X(x)$ describes the relative likelihood of X taking a given value. It is always non-negative and integrates to one over the entire domain.

The Cumulative Distribution Function (**CDF**)$F_X(x) = P(X \leq x)$ gives the probability that X does not exceed a threshold x:

$$F_X(x) \;=\; \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t. \tag{B.1}$$

In reliability analysis, the CDF is used directly to evaluate the probability of exceeding a critical value: $P(X > x_{\mathrm{cr}}) = 1 - F_X(x_{\mathrm{cr}})$ [2].

Two distributions are used in this work:

- **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{B.2}$$

- **Lognormal** distribution $\mathrm{LN}(\mu_{\ln}, \sigma_{\ln}^2)$, where $\ln X$ is normally distributed, ensuring $X > 0$:

$$f_X(x) = \frac{1}{x\,\sigma_{\ln}\sqrt{2\pi}} \exp\left[-\frac{(\ln x - \mu_{\ln})^2}{2\sigma_{\ln}^2}\right], \qquad x > 0. \tag{B.3}$$

The lognormal distribution is particularly relevant for material properties such as the Young modulus or permeability, which must remain strictly positive [6, 15, 1].

Statistical moments

The distribution of X can be summarised by its **statistical moments** [2], which quantify location, spread, asymmetry, and tail heaviness. In this work, the first two moments are prescribed as *input* to the stochastic analysis, while the higher moments are *computed as output* by the quadrature method (Appendix B.3).

Input moments. The **mean** μ_X and **variance** σ_X^2 (or equivalently the standard deviation $\sigma_X = \sqrt{\mathrm{Var}[X]}$) define the centre and spread of the distribution:

$$\mu_X = \int_{-\infty}^{+\infty} x\, f_X(x)\,\mathrm{d}x, \qquad \sigma_X^2 = \int_{-\infty}^{+\infty} (x-\mu_X)^2\, f_X(x)\,\mathrm{d}x. \tag{B.4}$$

Output moments. The **skewness** δ_3 and **kurtosis** δ_4 measure the asymmetry and tail heaviness of the response distribution:

$$\delta_3 = \frac{E[(X-\mu_X)^3]}{\sigma_X^3}, \qquad \delta_4 = \frac{E[(X-\mu_X)^4]}{\sigma_X^4}. \tag{B.5}$$

These four moments play a central role in the Gaussian Quadrature method (Appendix B.3), where the response PDF is reconstructed from the computed moments using Winterstein or Pearson approximations [14].

Coefficient of Variation (CoV)

The coefficient of variation quantifies the relative dispersion of X:

$$\mathrm{CoV}_X = \frac{\sigma_X}{\mu_X}. \tag{B.6}$$

It is a dimensionless measure that allows comparison of variability across parameters with different units and magnitudes. For concrete, typical values are $\mathrm{CoV}_E \approx 0.10$–$0.20$ for the Young's modulus [6] and $\mathrm{CoV}_K \approx 0.15$–$0.30$ for permeability [1].

B.1.2 Random fields and correlation structure

[REVIEW] This section is currently under development and will be refined as the study progresses.

Random fields

A random variable assigns a *single* uncertain value to a material property for the entire structure. In reality, concrete properties vary spatially due to heterogeneous microstructure, local variations in the water-to-cement ratio, and non-uniform compaction during casting [6]. A **random field** captures this spatial variability: it assigns an uncertain value to *every point* $\mathbf{x}$ in the domain, not just one value for the whole structure.

Formally, a random field is a collection of random variables indexed by position [15]:

$$H \;=\; \big\{H(\mathbf{x},\omega) : \mathbf{x} \in \mathcal{D},\ \omega \in \Omega\big\}, \tag{B.7}$$

where ω denotes one particular realisation drawn from the probability space. For a fixed ω, $H(\cdot,\omega)$ is a deterministic spatial function (one "map" of the property over the structure); for a fixed $\mathbf{x}$, $H(\mathbf{x},\cdot)$ is an ordinary random variable.

A **second-order** random field is fully characterised by two functions [15]:

- the **mean field** $\bar{H}(\mathbf{x}) = E[H(\mathbf{x})]$,
- the **covariance function** $C_H(\mathbf{x},\mathbf{y}) = E\big[\big(H(\mathbf{x}) - \bar{H}(\mathbf{x})\big)\big(H(\mathbf{y}) - \bar{H}(\mathbf{y})\big)\big]$.

The mean field gives the average value at each point; the covariance function measures how strongly the values at two points $\mathbf{x}$ and $\mathbf{y}$ are linked. When both depend only on the separation $\boldsymbol{\tau} = \mathbf{x} - \mathbf{y}$, the field is called **homogeneous** (stationary).

Correlation function and correlation length

When a material property (e.g. Young's modulus) is modelled as a random field, nearby points tend to have similar values while distant points are more different. The **autocorrelation function** $\rho_H(\boldsymbol{\tau})$ quantifies this similarity as a function of distance τ:

$$\rho_H(\boldsymbol{\tau}) \;=\; \frac{C_H(\boldsymbol{\tau})}{\sigma_H^2}, \qquad \rho_H(\mathbf{0}) = 1, \tag{B.8}$$

where $\sigma_H^2 = C_H(\mathbf{0})$ is the field variance. $\rho = 1$ means two points have identical values; $\rho \approx 0$ means they are nearly independent.

The rate at which ρ decays is controlled by the **correlation length** $L_c > 0$. Two common models for concrete [15, 8]:

$$\rho_H(\tau) = \exp\left(-\frac{\tau^2}{L_c^2}\right) \text{ (squared-exponential)}, \qquad \rho_H(\tau) = \exp\left(-\frac{|\tau|}{L_c}\right) \text{ (exponential)}. \tag{B.9}$$

Physically, L_c represents the distance beyond which material properties become essentially independent. A small L_c means the material varies rapidly in space (highly heterogeneous); a large L_c means properties change slowly (nearly uniform). This effect is illustrated in **??**.

Remark: In this work, L_c is not treated as a parameter to be varied. A single representative value is adopted from experimental data [6] to account for spatial heterogeneity in the Monte Carlo simulations (**??**). The objective is not to study the effect of L_c itself, but to compare the structural response *with* and *without* spatial variability.

B.1.3 Random field simulation

To use a random field in a finite element model, one must *generate* realisations that respect the prescribed covariance structure, then *assign* the values to elements or integration points. Several families of methods exist [8]:

- **Matrix decomposition (Cholesky).** The covariance matrix $\mathbf{C}$ is factorised as $\mathbf{C} = \mathbf{L}\mathbf{L}^T$ and a correlated realisation is obtained as $\mathbf{h} = \mathbf{L}\mathbf{z}$, where $\mathbf{z}$ is a vector of independent standard normal variates [8]. The method is exact but scales as $\mathcal{O}(n^3)$ in the number of nodes n, making it prohibitive for large meshes.

- **Spectral methods (FFT-based).** Realisations are generated by filtering white noise in the frequency domain using the spectral density corresponding to the target covariance [8]. FFT algorithms are computationally efficient on regular grids, but the method is limited to stationary fields on rectangular domains with periodic boundary conditions.

- **Local Average Subdivision (LAS).** LAS generates the random field by successively subdividing the domain and assigning local averages consistent with the target covariance at each level of refinement [8]. The method is computationally efficient and naturally produces spatially averaged values over elements, making it well-suited for direct use in finite element models. However, it is primarily designed for regular rectangular grids.

- **Series expansion methods.** Methods such as the Karhunen–Loève (KL) expansion parametrise the random field as a finite series of deterministic spatial functions multiplied by independent random variables [15, 6]. They provide a compact representation of the spatial variability but require solving a global eigenvalue problem on the finite element mesh, whose cost increases with mesh size and targeted accuracy.

- **Turning Bands Method (TBM).** The TBM reduces the d-dimensional simulation to a set of one-dimensional stationary processes along random lines (bands) passing through the domain [11, 8]. It requires neither a global matrix factorisation nor an eigenvalue solve, and operates directly on any mesh geometry.

In this work, the TBM method, described in detail in Appendix B.2.4, is adopted because it is the method implemented in the Cast3M `ALEA` operator.

B.1.4 Summary: random variable versus random field

[REVIEW] This section is under review Table B.1 summarises the key differences between the two levels of stochastic description and their implications for the numerical methods used in this work.

Table B.1: Comparison of random variable and random field stochastic descriptions.

Aspect	**Random variable**	**Random field**
Spatial variation	None (uniform)	Yes (spatially heterogeneous)
Characterisation	PDF: mean μ, std σ	Mean field $\bar{H}(\mathbf{x})$, covariance $C_H(\mathbf{x}, \mathbf{y})$
Key parameter	CoV	Correlation length L_c
Stochastic level	First-order	Second-order
Simulation method	Gaussian Quadrature	TBM + Monte Carlo
Cast3M operator	`QUADRATU`	`ALEA`
Computational cost	Low (3–7 runs per variable)	High (hundreds to thousands of runs)

B.2 Monte Carlo simulation framework

B.2.1 Principle

Monte Carlo simulation (MCS) estimates the effect of uncertain inputs by running the same deterministic model many times, each time with a different random sample [8]. Each trial consists of three steps:

1. Generate a realisation of the random input (random variable or random field) according to its prescribed distribution.
2. Run the deterministic model (here, the THM finite element solver in Cast3M) to obtain the response.
3. Collect the result; repeat until the output statistics are stable.

Because no assumption is made on the shape of the response, MCS can estimate the *entire* output distribution—unlike linearisation methods such as FORM/SORM [8].

B.2.2 Convergence

The probability of failure is estimated as the fraction of realisations where the limit state is violated:

$$\hat{P}_f = \frac{N_f}{n}, \tag{B.10}$$

where the hat denotes that $\hat{P}_f$ is an estimate of the true P_f, which improves as n increases, and N_f is the number of failures out of n total runs. The more runs, the more accurate the estimate—but the error only decreases as $1/\sqrt{n}$. To halve the error, one must quadruple the number of runs.

At 95% confidence, the required number of runs is [8]:

$$n \;\geq\; \frac{4\, p_f (1 - p_f)}{e^2}, \tag{B.11}$$

where e is the acceptable absolute error. For a small failure probability ($p_f \approx 10^{-3}$) with 10% relative accuracy, this gives $n \approx 400{,}000$ runs. When a single THM analysis takes several minutes, this cost becomes prohibitive—motivating the Gaussian Quadrature method (Appendix B.3) as a faster alternative.

B.2.3 Transformation of Lognormal to Gaussian distribution

In practical finite element implementations using the Cast3M software, material parameters such as Young's modulus and permeability must remain strictly positive. However, the built-in random field operator `ALEA` generates Gaussian (normal) distributions, necessitating a mathematical transformation [15].

A **lognormal random field** $H(\mathbf{x})$ is therefore adopted: by definition, $\ln H(\mathbf{x})$ is a Gaussian random field, which guarantees $H(\mathbf{x}) > 0$ at every point [6]. The moments of $H(\mathbf{x})$ are related to those of the underlying Gaussian field $G(\mathbf{x}) \sim \mathcal{N}(\mu_{\ln}, \sigma_{\ln}^2)$ by [15]:

$$\begin{cases} \mu_X &= \exp\left(\mu_{\ln} + \dfrac{\sigma_{\ln}^2}{2}\right) \\ \sigma_X^2 &= [\exp(\sigma_{\ln}^2) - 1] \exp(2\mu_{\ln} + \sigma_{\ln}^2) \end{cases} \tag{B.12}$$

where μ_X and σ_X are the physical mean and standard deviation of the material property. Inverting Equation (B.12) yields the Gaussian parameters as functions of the physical moments:

$$\sigma_{\ln}^2 = \ln\bigl(1 + CoV_X^2\bigr)\,, \qquad \mu_{\ln} = \ln(\mu_X) - \tfrac{1}{2}\,\sigma_{\ln}^2, \tag{B.13}$$

where $CoV_X = \sigma_X/\mu_X$ is the coefficient of variation.

The standard computational procedure in Cast3M `ALEA` then consists of two steps:

1. Generate a Gaussian random field $G(\mathbf{x}) \sim \mathcal{N}(\mu_{\ln}, \sigma_{\ln}^2)$ using the Turning Band Method (see Appendix B.2.4), with parameters computed from Equation (B.13).

2. Transform pointwise to obtain the lognormal field:

$$H(\mathbf{x}) \;=\; \exp\bigl(G(\mathbf{x})\bigr), \tag{B.14}$$

   which guarantees $H(\mathbf{x}) > 0$ and preserves the correlation structure characterised by L_c (see Appendix B.1.2).

This pointwise transformation is exact: the autocorrelation structure of the underlying Gaussian field is preserved in the lognormal field, so the spatial variability characterised by the correlation length L_c remains physically meaningful after transformation [6].

B.2.4 Turning Bands Method (ALEA)

Principle The Turning Bands Method (TBM), introduced by [11], simplifies the generation of a 2-D or 3-D random field by reducing it to a set of 1-D problems.

The idea (**??**): draw L lines through the domain in different directions. Along each line, simulate an independent 1-D random process. To obtain the field value at any point $\mathbf{x}$, project it onto each line and average the results:

$$Z(\mathbf{x}) \;=\; \frac{1}{\sqrt{L}} \sum_{i=1}^{L} Z_i(\mathbf{x} \cdot \mathbf{u}_i)\,, \tag{B.15}$$

where $\mathbf{u}_i$ is the direction of the i-th line and Z_i is the 1-D process along it. The 1-D processes are constructed so that the resulting field reproduces the prescribed covariance structure (Appendix B.1.2). In practice, $L = 16$ lines are sufficient for isotropic fields in two dimensions [8].

Implementation in Cast3M The TBM is natively implemented in the Cast3M `ALEA` operator, which takes as input the Gaussian parameters $(\mu_{\ln}, \sigma_{\ln})$ and the correlation length L_c, and returns the simulated field directly as nodal values on the mesh. The lognormal transformation and assignment to `MATE` follow the procedure described in Appendix B.2.3.

B.2.5 Integration of spatial variability into the THM model

This section is currently under development and will be refined as the study progresses.

Once the lognormal random field $H(\mathbf{x})$ has been constructed via the Turning Band Method (see Appendix B.2.4), it is propagated through the nonlinear THM model using Monte Carlo simulation. Given the computational cost of each THM evaluation, the iterative procedure is fully automated in Cast3M using the `REPETER` operator.

Simulation loop and dynamic material updating

For a predefined number of simulations N_{simu}, the global algorithm performs the following sequential steps within each loop:

1. **Random Field Generation:** A new statistically independent realization of the random field is generated using the `ALEA` operator. For instance, the spatial distribu-

tion of the Young's modulus $\mathbf{E}(x)$ is recalculated based on the assigned log-normal parameters (mean, standard deviation, and correlation length).

2. **Material Updating:** The macroscopic properties of the finite element model are updated. The newly generated random field is projected onto the mesh to form the updated material stiffness matrix using the `MATE` operator.

3. **Non-linear Resolution:** The `PASAPAS` procedure is invoked to solve the multi-physics THM problem from $t = 0$ to the final time t_{fin}, utilizing the updated heterogeneous material properties.

B.3 Gaussian Quadrature framework

Unlike Monte Carlo, which runs thousands of simulations, Gaussian Quadrature evaluates the model at a small number of strategically chosen points and reconstructs the output statistics from a weighted sum [**baldeweck1999**, 14]. The method has three steps, corresponding to three Cast3M operators:

1. `QUADRATU`: compute the evaluation points and weights for each random variable.
2. `PARASTAT`: run the model at each point combination and compute the first four statistical moments (mean, standard deviation, skewness, kurtosis).
3. `PROBDENS`: reconstruct the PDF from these moments and estimate P_f and β.

B.3.1 Step 1: Quadrature points and weights (`QUADRATU`)

Computing any statistical moment of the output $G = g(\mathbf{X})$ requires evaluating an integral:

$$I = \int_{\mathcal{D}} g(x)\, W(x)\, \mathrm{d}x, \tag{B.16}$$

where $W(x) = f_X(x)$ is the PDF of the input variable. Gaussian quadrature approximates this by a weighted sum over K evaluation points [**baldeweck1999**]:

$$I \approx \sum_{k=1}^{K} w_k\, g(x_k), \tag{B.17}$$

where $\{x_k\}$ are the *abscissas* (evaluation points) and $\{w_k\}$ the corresponding *weights*. The optimal abscissas are the roots of orthogonal polynomials associated with W, and a K-point scheme integrates exactly any polynomial of degree up to $2K - 1$ [14].

`QUADRATU` automatically computes $\{x_k, w_k\}$ for any prescribed distribution (Normal, Log-normal, Uniform, etc.). The user provides only the distribution type and its parameters.

B.3.2 Step 2: Statistical moments (`PARASTAT`)

For N independent random variables with K points each, the quadrature is applied to each variable separately and combined as a tensor product [14]:

$$\mu_q^{(0)} \approx \sum_{k_1=1}^{K_1} \cdots \sum_{k_N=1}^{K_N} w_{k_1} \cdots w_{k_N} \left[g(x_{k_1}, \ldots, x_{k_N}) \right]^q, \tag{B.18}$$

requiring K^N model evaluations in total. For example, with $N = 2$ variables and $K = 3$ points each: $3^2 = 9$ runs.

`PARASTAT` collects these evaluations and returns the first four moments: mean μ, standard deviation σ, skewness $\delta_3 = \mu_3/\sigma^3$, and kurtosis $\delta_4 = \mu_4/\sigma^4$.

B.3.3 Step 3: PDF reconstruction and reliability (`PROBDENS`)

Once the four moments are known, `PROBDENS` reconstructs the PDF using *three independent methods simultaneously* [14]:

- **Winterstein approximation:** expresses G as a Hermite polynomial transformation of a standard normal variable. Simplest but least accurate of the three methods.
- **Pearson's method:** selects the best-fit distribution family (normal, beta, gamma, etc.) from the moment values. Most accurate, but does not always provide a result.
- **Exponential of polynomials ($n = 4$):** models the PDF as $\tilde{f}_G = c\,\exp(Q)$ where Q is a polynomial fitted to the moments. Always provides accurate results.

Each method independently yields a probability of failure P_f and reliability index β. When the three results are consistent with each other, they can be accepted with confidence [14]:

$$P_f = \tilde{F}_G(0), \qquad \beta = -\Phi^{-1}(P_f), \tag{B.19}$$

where $\tilde{F}_G$ is the reconstructed CDF and Φ^{-1} the inverse standard normal CDF. The method provides accurate results for $\beta \lesssim 5$ [14].

B.3.4 Computational cost

The total number of model evaluations is K^N, which grows exponentially with N. For the present study with $N = 2$ and $K = 3$, this gives only 9 runs—compared to thousands for Monte Carlo. However, for large N (e.g. $3^{10} = 59{,}049$ runs), the cost becomes comparable to Monte Carlo. This *curse of dimensionality* limits the method to problems with a small number of random variables ($N \lesssim 10$–15) [14].

B.4 Sensitivity indicators and ranking methods

B.4.1 Variance decomposition

For a model $G = g(X_1, \ldots, X_N)$ with N independent random inputs, Hoeffding's decomposition [**sobol1993**] expresses G as a sum of functions of increasing dimension:

$$G \;=\; g_0 \;+\; \sum_{i=1}^{N} g_i(X_i) \;+\; \sum_{i<j} g_{ij}(X_i, X_j) \;+\; \cdots \;+\; g_{1,\ldots,N}(X_1, \ldots, X_N), \tag{B.20}$$

where $g_0 = \mathbb{E}[G]$ is a constant and each term has zero mean. Taking the variance of both sides yields:

$$\mathrm{Var}(G) \;=\; \sum_{i=1}^{N} V_i \;+\; \sum_{i<j} V_{ij} \;+\; \cdots \;+\; V_{1,\ldots,N}, \tag{B.21}$$

where $V_i = \mathrm{Var}[g_i(X_i)]$ is the first-order variance (effect of X_i alone) and $V_{ij} = \mathrm{Var}[g_{ij}(X_i, X_j)]$ is the second-order variance (interaction between X_i and X_j).

B.4.2 Variance-based sensitivity: Sobol indices

Dividing each term in Equation (B.21) by the total variance defines the *Sobol indices* [**sobol1993**]:

$$S_i \;=\; \frac{V_i}{\mathrm{Var}(G)}, \qquad S_{ij} \;=\; \frac{V_{ij}}{\mathrm{Var}(G)}, \tag{B.22}$$

where:

- S_i is the **first-order Sobol index**: fraction of the output variance due to X_i alone.

- S_{ij} is the **second-order Sobol index**: fraction due to the interaction between X_i and X_j, beyond their individual effects.

The **total-order Sobol index** captures the full effect of X_i, including all interactions:

$$S_{T_i} = S_i + \sum_{j \neq i} S_{ij} + \cdots \tag{B.23}$$

By construction, the first-order indices sum to at most one ($\sum_i S_i \leq 1$), and the total-order indices sum to at least one ($\sum_i S_{T_i} \geq 1$). When the model is additive (no interactions), $S_{ij} = 0$ for all pairs and $S_i = S_{T_i}$.

The first-order index S_i can be expressed as a conditional variance:

$$S_i = \frac{\mathrm{Var}_{X_i}\left[\mathbb{E}_{X_{\sim i}}(G \mid X_i)\right]}{\mathrm{Var}(G)}, \tag{B.24}$$

where $X_{\sim i}$ denotes all variables except X_i. Intuitively, $\mathbb{E}(G \mid X_i)$ averages out all other variables: if this conditional mean varies a lot with X_i, then X_i is influential.

A variable with a large S_i (or S_{T_i}) is a *dominant parameter*: it drives most of the output uncertainty and should be characterised with care. A variable with a small index could be treated as deterministic without significant loss of accuracy.

B.4.3 Practical sensitivity approximation

Computing exact Sobol indices requires a large number of model evaluations. A simpler approach, used by the the expansion of Englishquadrature method (Appendix B.3), estimates the contribution of each variable one at a time:

1. Fix all variables at their mean except X_i.
2. Run the model at each quadrature point of X_i to obtain the marginal variance $\hat{\sigma}_i^2$.
3. Repeat for every variable.

The **contribution** of X_i to the total output variability is then:

$$C_i = \frac{\hat{\sigma}_i^2}{\sum_{j=1}^{N} \hat{\sigma}_j^2} \times 100\,\% \approx S_i \quad \text{when } S_{ij} \approx 0. \tag{B.25}$$

A variable with $C_i \approx 0\,\%$ has negligible influence and could be treated as deterministic. This approximation is valid when variables are uncorrelated and interaction effects are small. The detailed results are presented in **??**.

B.5 Probability of failure and reliability index

B.5.1 Limit state function

In the sensitivity analysis (Section 5.5), $G = g(X_1, \ldots, X_N)$ denoted the model output without any notion of safety or failure. In reliability analysis, the same symbol G is given an additional meaning: its *sign* now defines whether the structure is safe or has failed.

Specifically, G is rewritten as a **performance function** (also called limit-state function or safety margin) [2]. A simple example: if R is the material resistance and S is the applied load, then $G = R - S$. The structure is safe when $G > 0$ and fails when $G \leq 0$:

$$G(\mathbf{X}) > 0 \;\Rightarrow\; \text{safety domain } D_s, \qquad G(\mathbf{X}) \leq 0 \;\Rightarrow\; \text{failure domain } D_f. \tag{B.26}$$

In the simple case of a resistance R and a load effect S, the performance function writes $G(R, S) = R - S$ [2].

B.5.2 Probability of failure

The probability of failure is defined as [2]:

$$P_f \;=\; \text{Prob}\big[G(\mathbf{X}) \leq 0\big] \;=\; \int_{D_f} f_{\mathbf{X}}(\mathbf{x})\,\mathrm{d}\mathbf{x}, \tag{B.27}$$

where $f_{\mathbf{X}}$ is the joint PDF of the random input vector. This integral is generally intractable for complex systems, which motivates the use of reliability indices.

B.5.3 Reliability indices

Given P_f from Equation (B.27), the *generalized reliability index* is defined as [14]:

$$\hat{\beta} \;=\; -\Phi^{-1}\big(\hat{P}_f\big), \tag{B.28}$$

where Φ is the standard normal CDF. This definition applies to both the quadrature method (where $\hat{P}_f$ is obtained from the reconstructed CDF, Appendix B.3.3) and Monte Carlo simulation (where $\hat{P}_f$ is the empirical failure ratio, Appendix B.2).

Remark: Classical reliability theory also defines the Cornell index $\beta_C = \mu_G/\sigma_G$ and the Hasofer–Lind index β_{HL} (minimum distance to the failure surface in the standard normal space) [2, 1]. The Cornell index is exact only for linear limit states; the Hasofer–Lind index is used in FORM/SORM methods. Neither is employed in this work; for details, the reader is referred to [2, 1].

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EDUCATION

Université Grenoble Alpes **Sep 2025 - Present**

Master 2: Geomechanics, Civil Engineering & Risks *Grenoble, France*

- **M2 Semester 1 Grade:** 13.47/20 (Admis).
- **Key Coursework:** Numerical Methods for Nonlinear Mechanics, Durability and Vulnerability of Structures (15.8/20), Quantitative Image Analysis for Mechanics (16/20).
- **Thesis:** Numerical and experimental study of the THM behavior of concrete under severe accident conditions using a poro-mechanical approach.

Ho Chi Minh University of Technology **Sep 2019 - Jun 2024**

Engineering Degree in Building & Energy (PFIEV Program) *HCM City, Vietnam*

- **GPA:** 8.06/10 (3.5/4.0) | **Rank:** 3/12 (Excellence Program).
- **Honors:** University Semester Scholarships for 4 consecutive years (2021-2024)

RESEARCH & PROFESSIONAL EXPERIENCE

M2 Research Intern / Thesis Student **Feb 2026 - Present**

3SR Laboratory, Université Grenoble Alpes *Grenoble, France*

- **Repository:** github.com/tankiet1907/cast3m-m2internship
- Investigating the thermo-hydro-mechanical (THM) behavior of concrete under severe accident conditions using a poro-mechanical approach.
- Developing and implementing numerical behavior models within the **Cast3M** finite element code using Gibiane.
- Utilizing probabilistic and statistical methods to conduct sensitivity analyses and optimize THM coupling parameters.
- Correlating numerical simulations with experimental data to validate predictive models for structural integrity.

Structure Engineer (Full Time) **Aug 2024 - Aug 2025**

GBC Engineers Vietnam *Ho Chi Minh City, Vietnam*

- Prepared Finite Element Models (FEM) and structural models for rigorous mechanical analysis.
- Delivered structural calculations, analysis reports, and correspondence ensuring compliance with international standards (Eurocode, TCVN).

Site Engineer (Internship) **Jun 2023 - Aug 2023**
Unicons Construction (Member of Coteccons Group) *Binh Duong, Vietnam*

- **Project:** LEGO Manufacturing Vietnam (Phase 1) – Moulding Building.
- Supervised structural fire protection implementations (Fire Protection Boards, Intumescent Paint, Concrete-encasement) to meet rigorous 120-minute Fire Protection Ratings.
- Optimized the execution schedule by transitioning the concrete encasement casting for columns from vertical to horizontal pouring.
- Conducted Quality Control per EC2: inspected steel frame erection via total station, verified tightening forces for Snug and Bearing bolts, and supervised Sika grouting and composite decking.

PROJECTS

Scientific Research: Fire-Resistant Steel Structures **Sep 2023 - Dec 2023**
Ho Chi Minh City University of Technology *Ho Chi Minh City, Vietnam*

- Evaluated thermo-mechanical properties of steel at high temperatures per EUROCODE specifications.
- Designed and verified fire-resistant structural elements (columns and beams) using advanced material models.

CERTIFICATIONS & AWARDS

- **Third Prize,** 36^{th} **Loa Thanh Award (2024):** National award for the most prestigious graduation projects in civil engineering.
- **Third Prize in Physics (2018):** City Level Excellent Student Competition (Ho Chi Minh City).
- **Certificate of Merit:** Recognized as a "Very Good Student" for the 2021-2022 academic year.

SKILLS & COMPETENCIES

Technical: Cast3M (Advanced), Finite Element Analysis (FEA), THM Coupling, Probabilistic Sensitivity Analysis.

Software: LaTeX, Linux Mint, VS Code, Revit Structure, AutoCAD, ETABS, SAP2000, MS Office.

Languages: Vietnamese (Native), English (Advanced - TOEIC: 935/990), French (Intermediate - TCF TP: 347/699, B1).

EXTRACURRICULAR ACTIVITIES

- **Main Vocal:** Tet Festival of the Vietnamese Community in Grenoble. 2026
- **Main Vocal:** PFIEV Freshman Welcome Party (2023) and Journée de la Culture Française (2022). 2022 - 2023
- **Volunteer:** Supporting COVID-19 prevention work.
- **Security Support:** Admissions Consulting Day (2022) and University Entrance Exam Support Program (2020). 2020 - 2022.